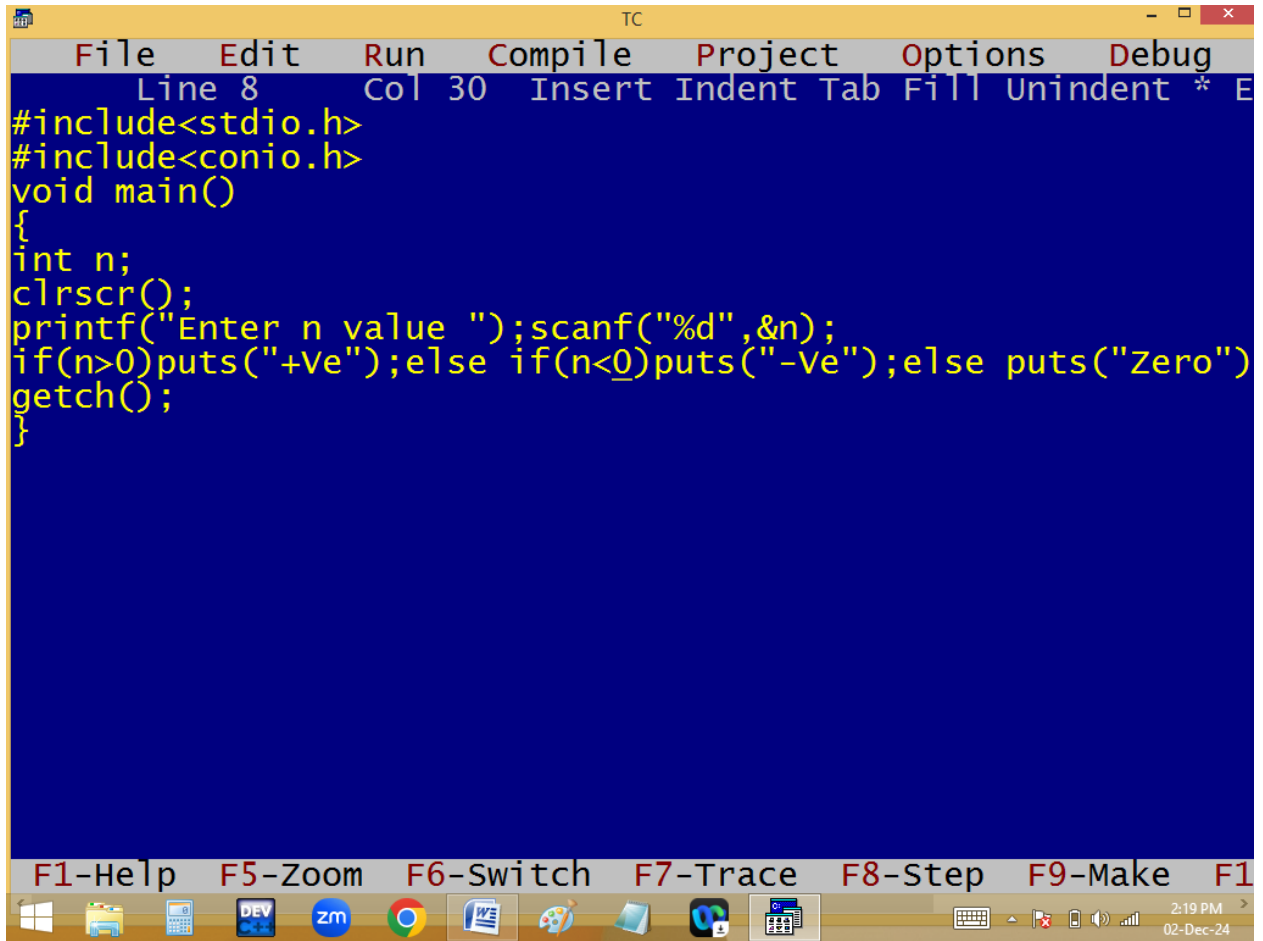


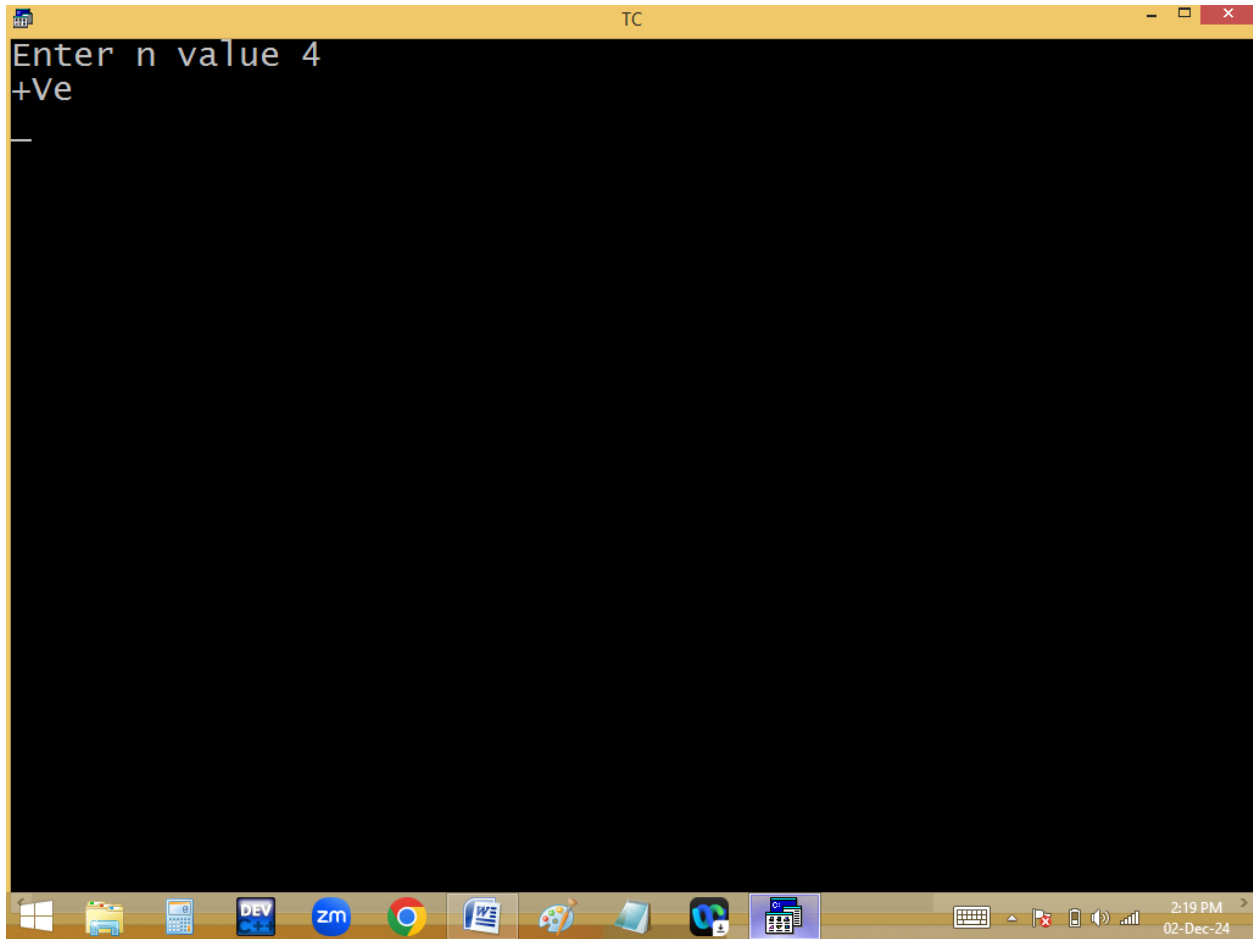
Finding +Ve / -Ve / 0

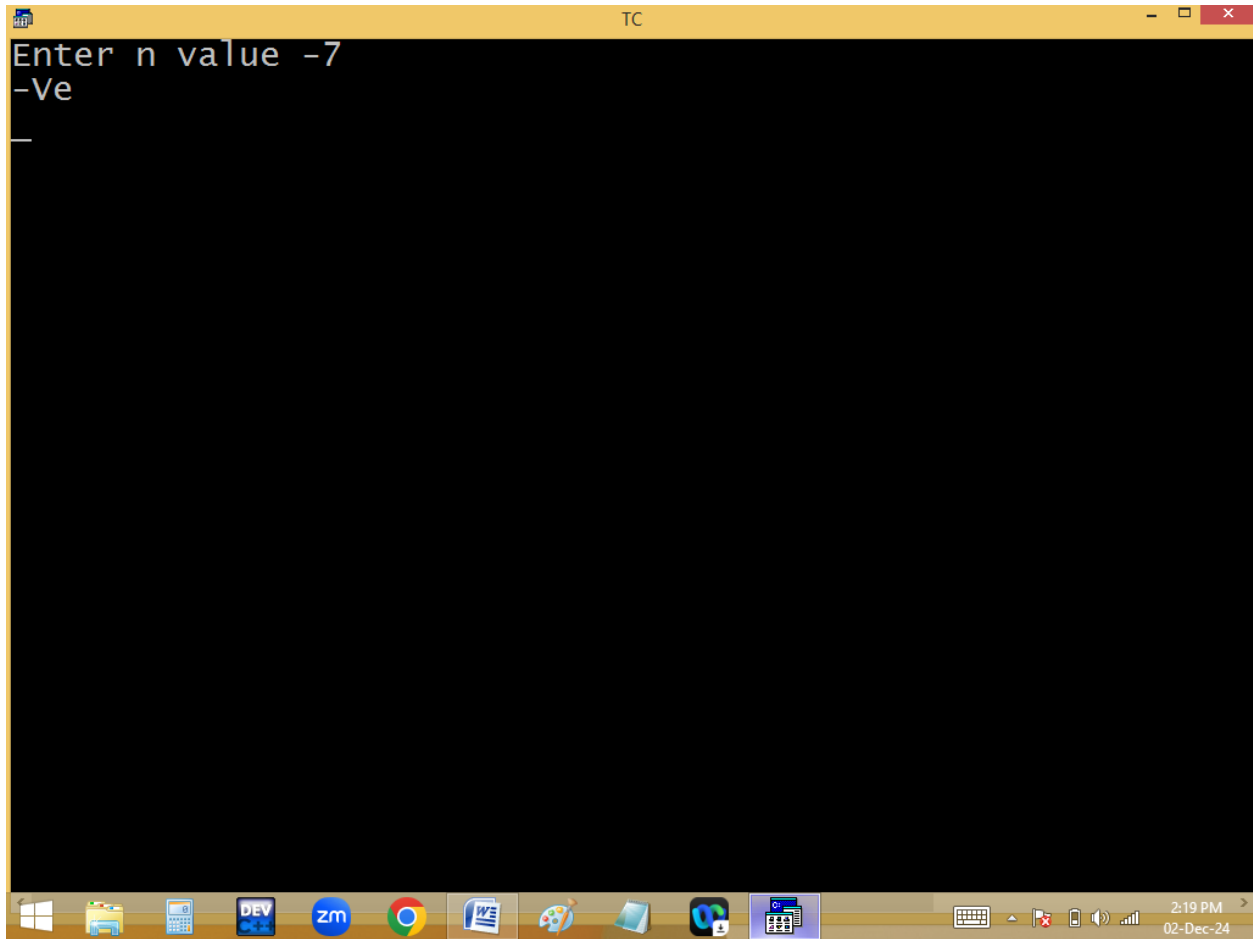


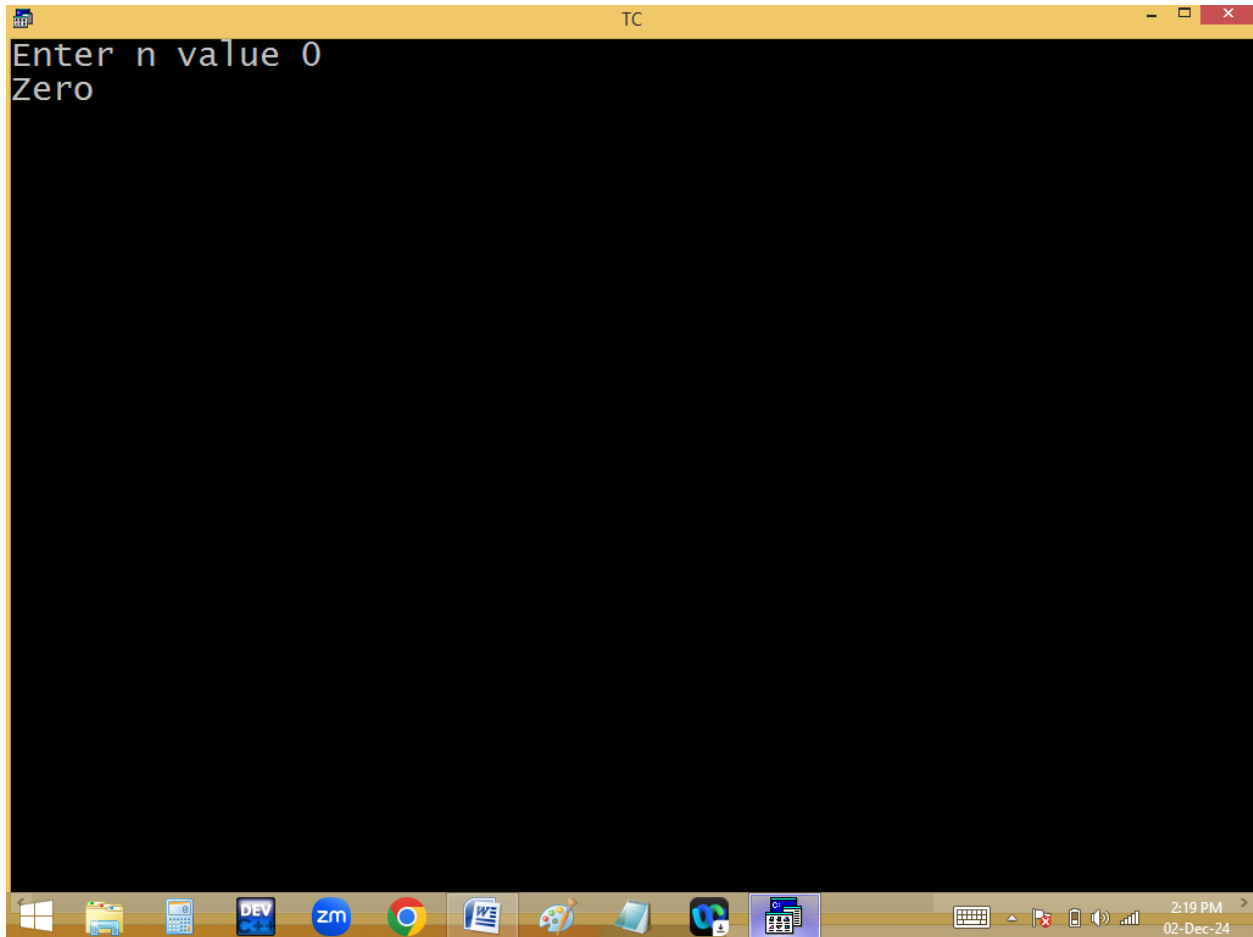
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 8 Col 30 Insert Indent Tab Fill Unindent * E". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter n value ");scanf("%d",&n);
if(n>0)puts("+Ve");else if(n<0)puts("-Ve");else puts("Zero")
getch();
}
```

The bottom status bar shows function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. The Windows taskbar at the bottom includes icons for File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Notepad, Paint, and the Start menu. The system clock shows 2:19 PM on 02-Dec-24.







Finding no of days in given month.

1, 3, 5, 7, 8, 10,12 – 31 days

2 – 28/29 days

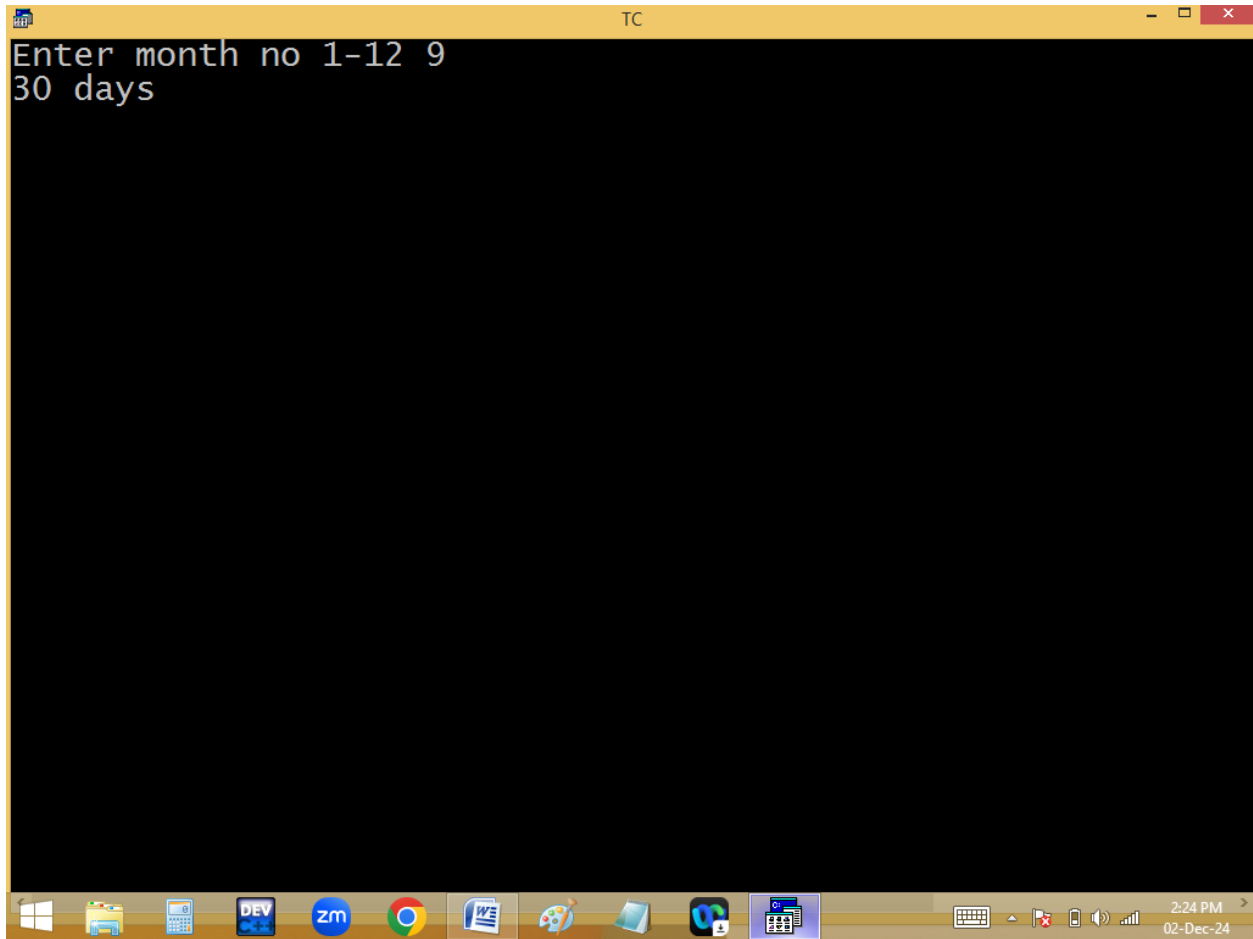
4, 6, 9, 11 – 30 days

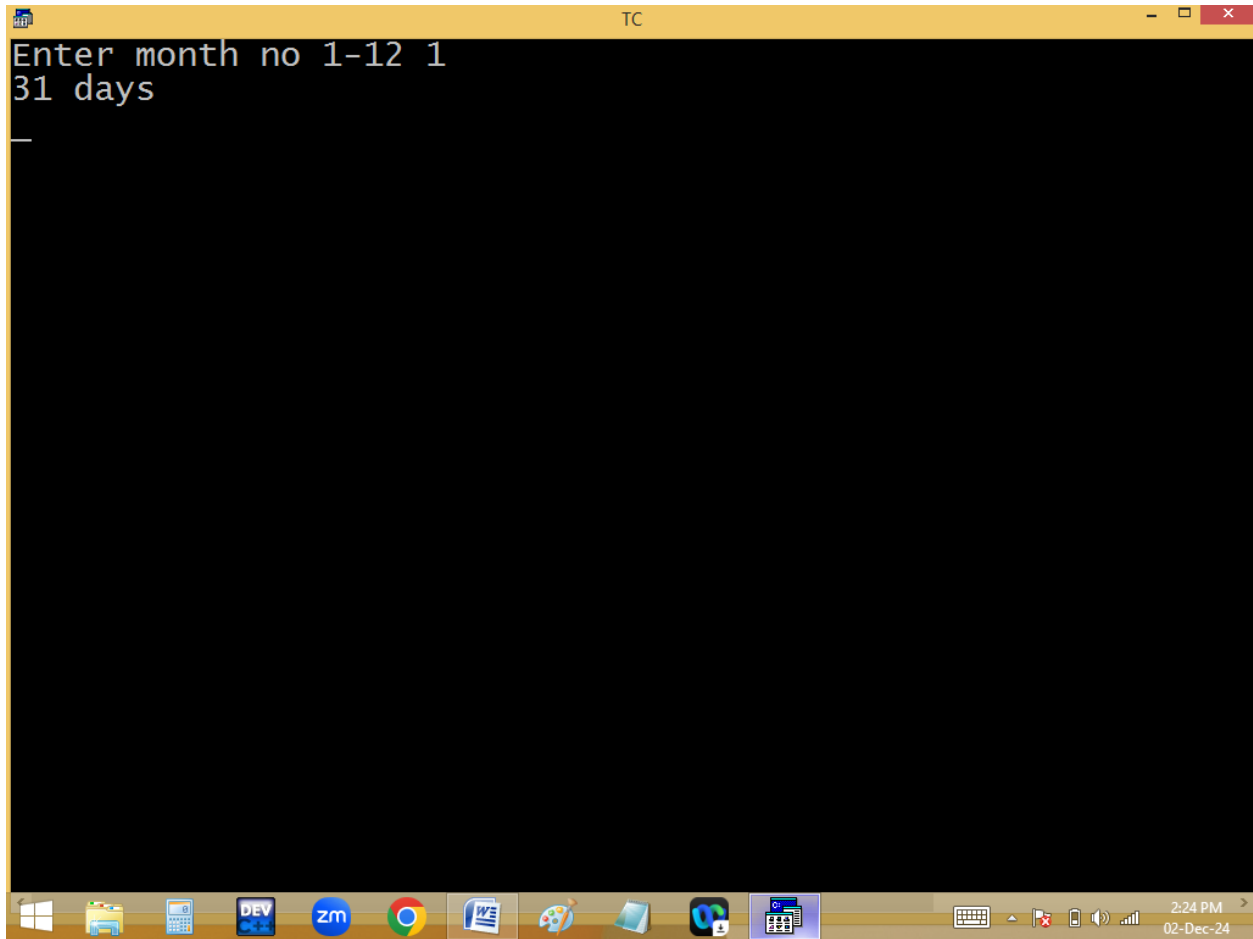
Others – invalid month no

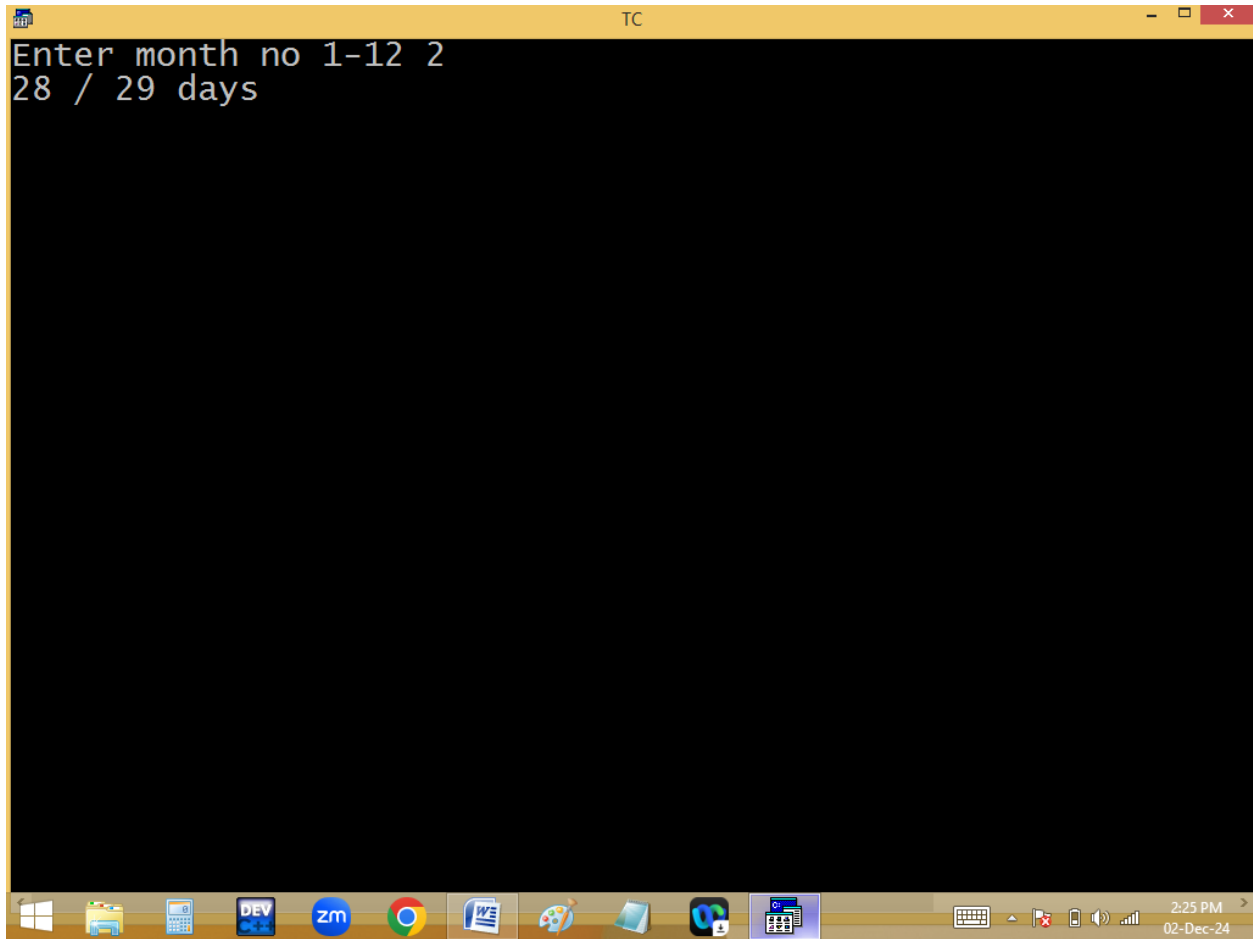
```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 22 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter month no 1-12 ");scanf("%d",&n);
if(n==1||n==3||n==5||n==7||n==8||n==10||n==12)puts("31 days"
else if(n==2)puts("28 / 29 days");
else if(n==4 || n==6 || n==9 || n==11)puts("30 days");
else puts("Invalid Month no");
getch();
}
```

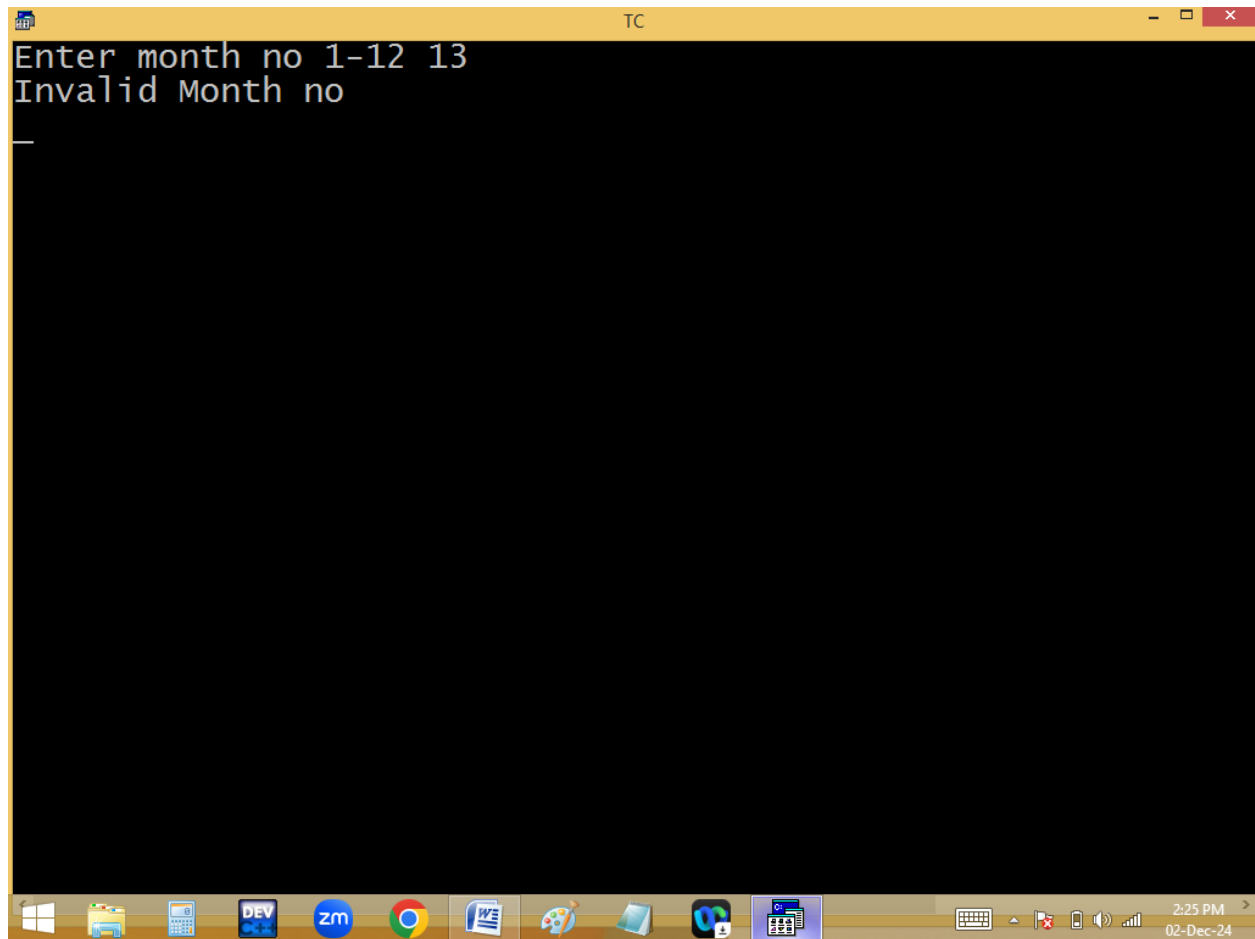
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

2:24 PM 02-Dec-24

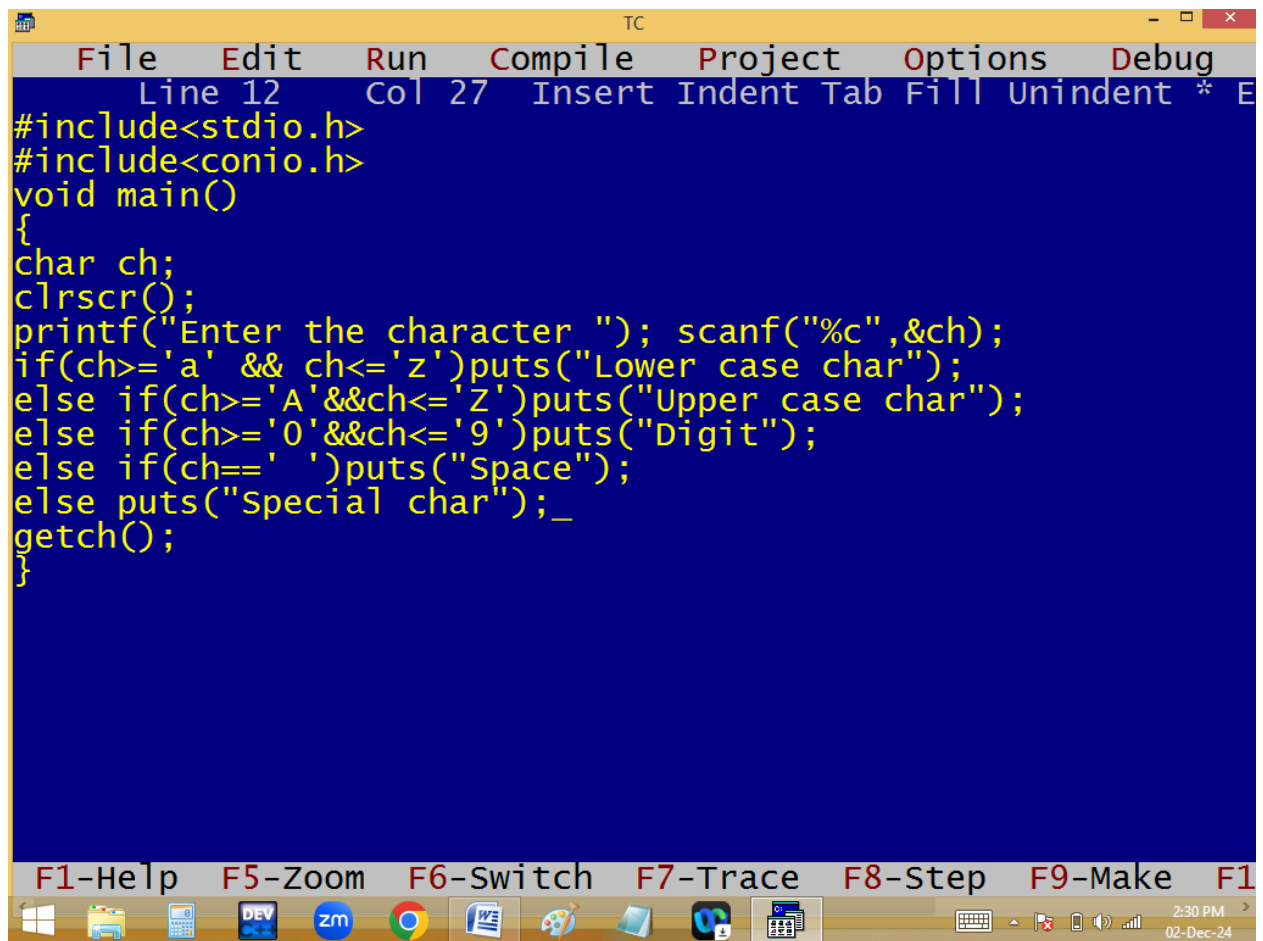








Find lower / upper / digit / space / special char?

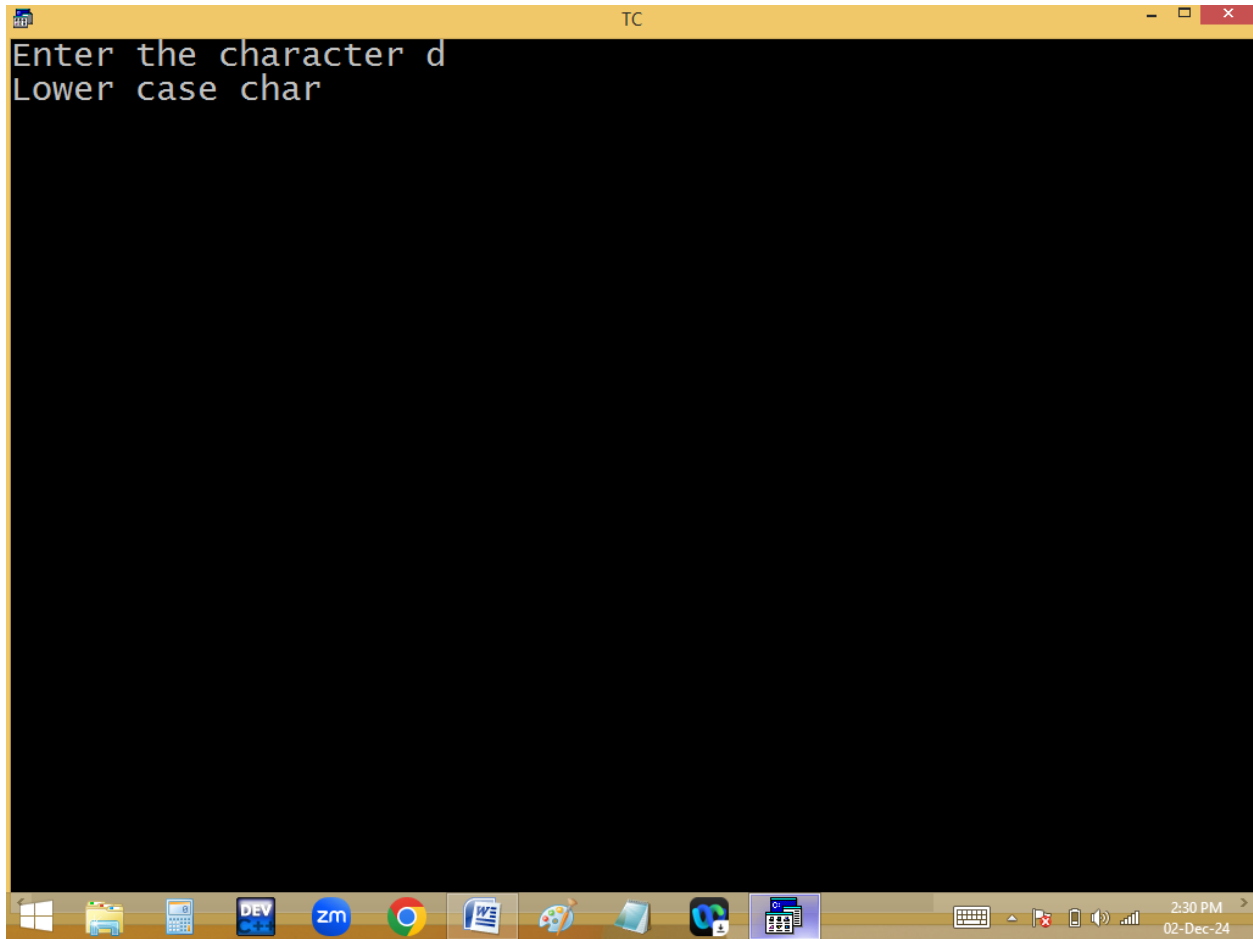


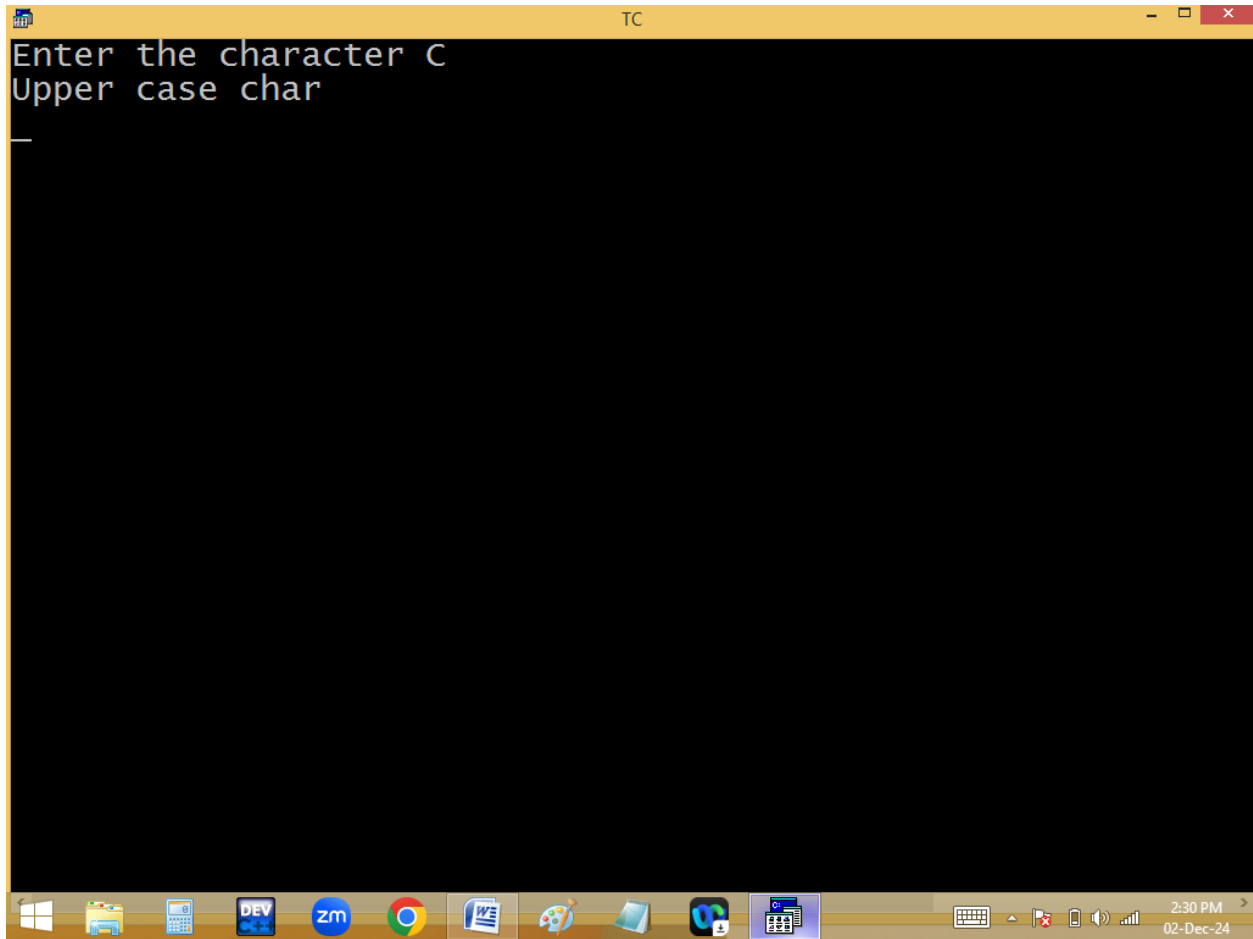
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 12", "Col 27", and "Insert" mode. The code is written in yellow text on a blue background. The code defines a function `main()` that prompts the user to enter a character and then classifies it as a lower case character, upper case character, digit, space, or special character. The taskbar at the bottom shows various application icons and the system clock indicating 2:30 PM on 02-Dec-24.

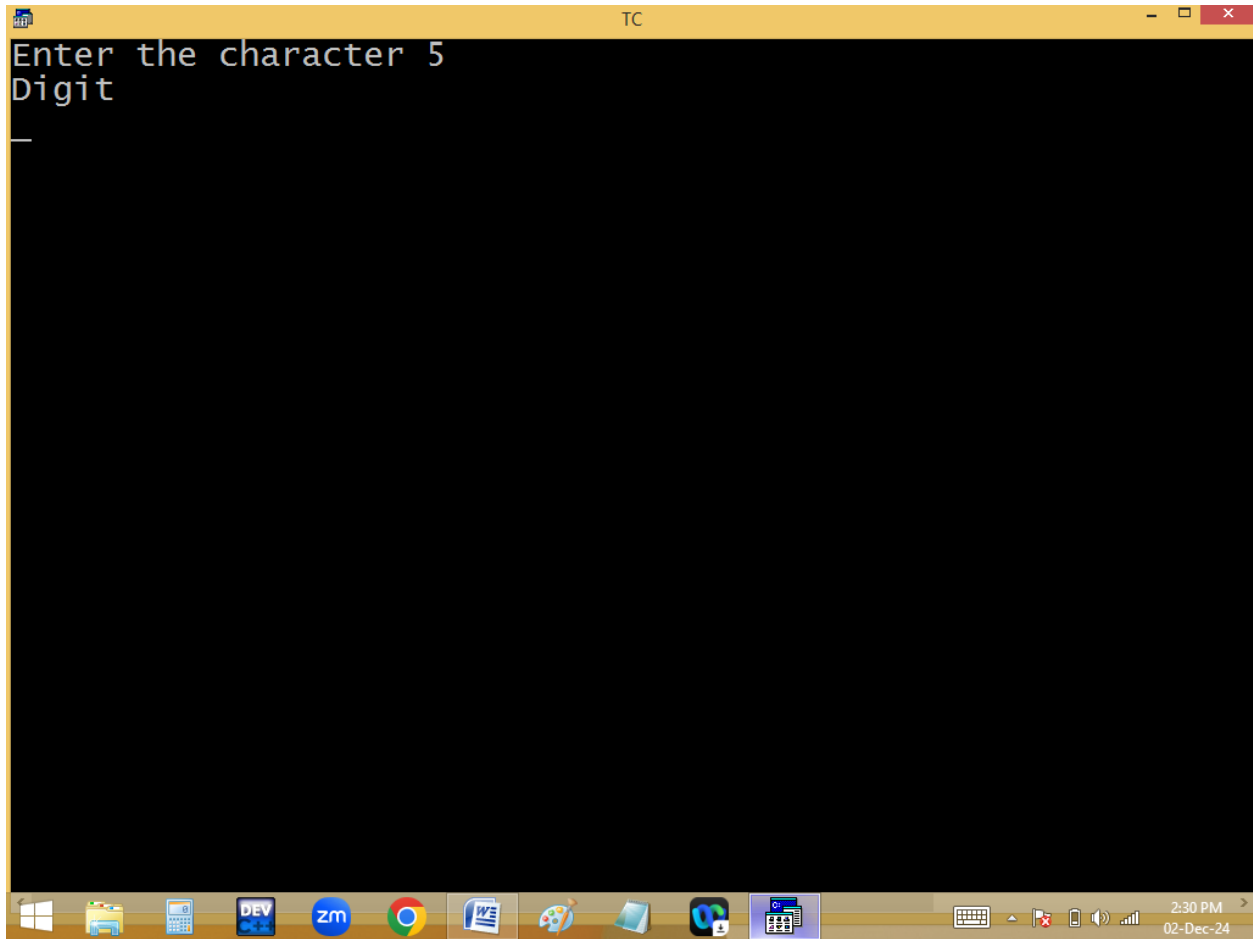
```
File Edit Run Compile Project Options Debug
Line 12 Col 27 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(ch>='a' && ch<='z')puts("Lower case char");
else if(ch>='A'&&ch<='Z')puts("Upper case char");
else if(ch>='0'&&ch<='9')puts("Digit");
else if(ch==' ')puts("Space");
else puts("Special char");_
getch();
}
```

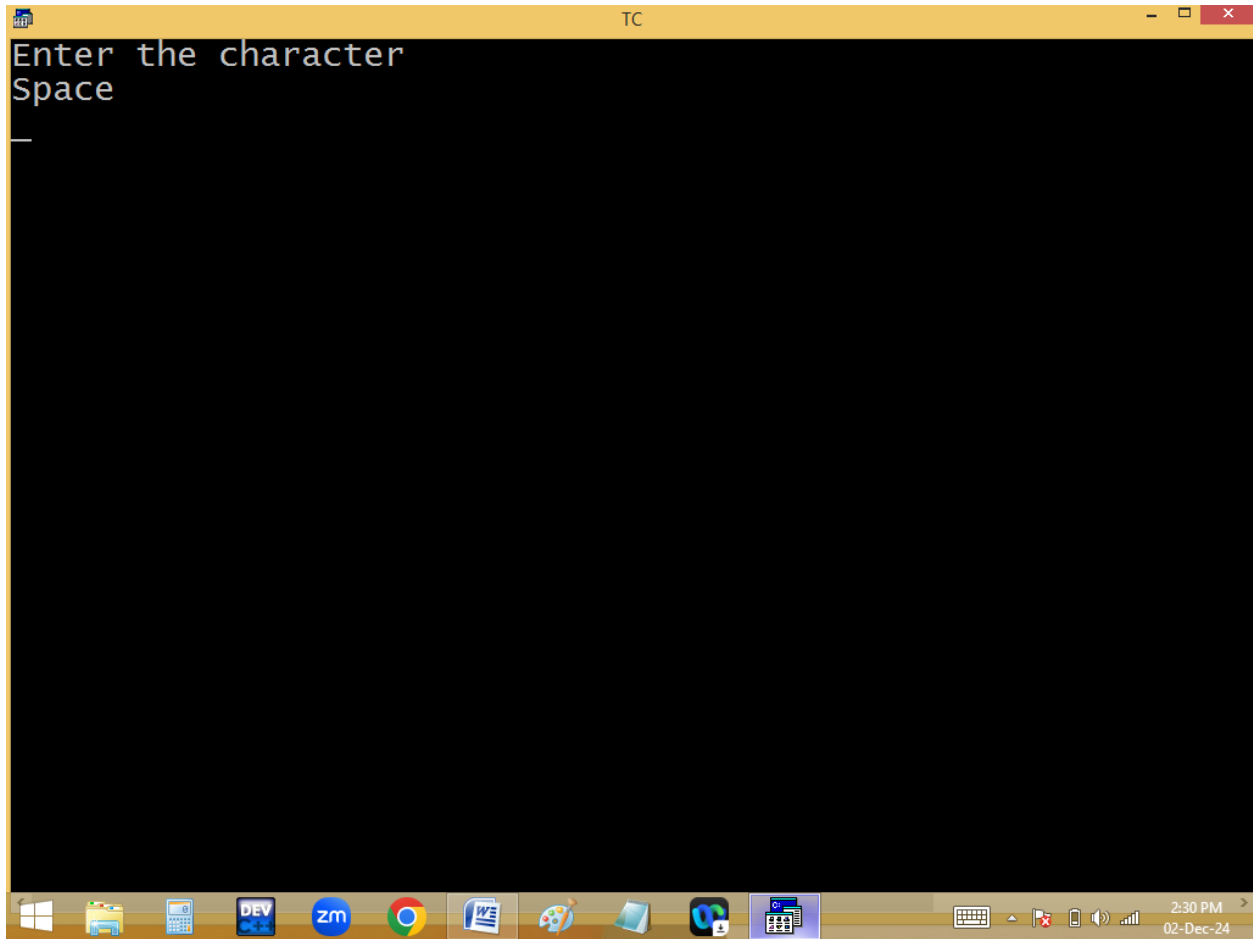
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Exit

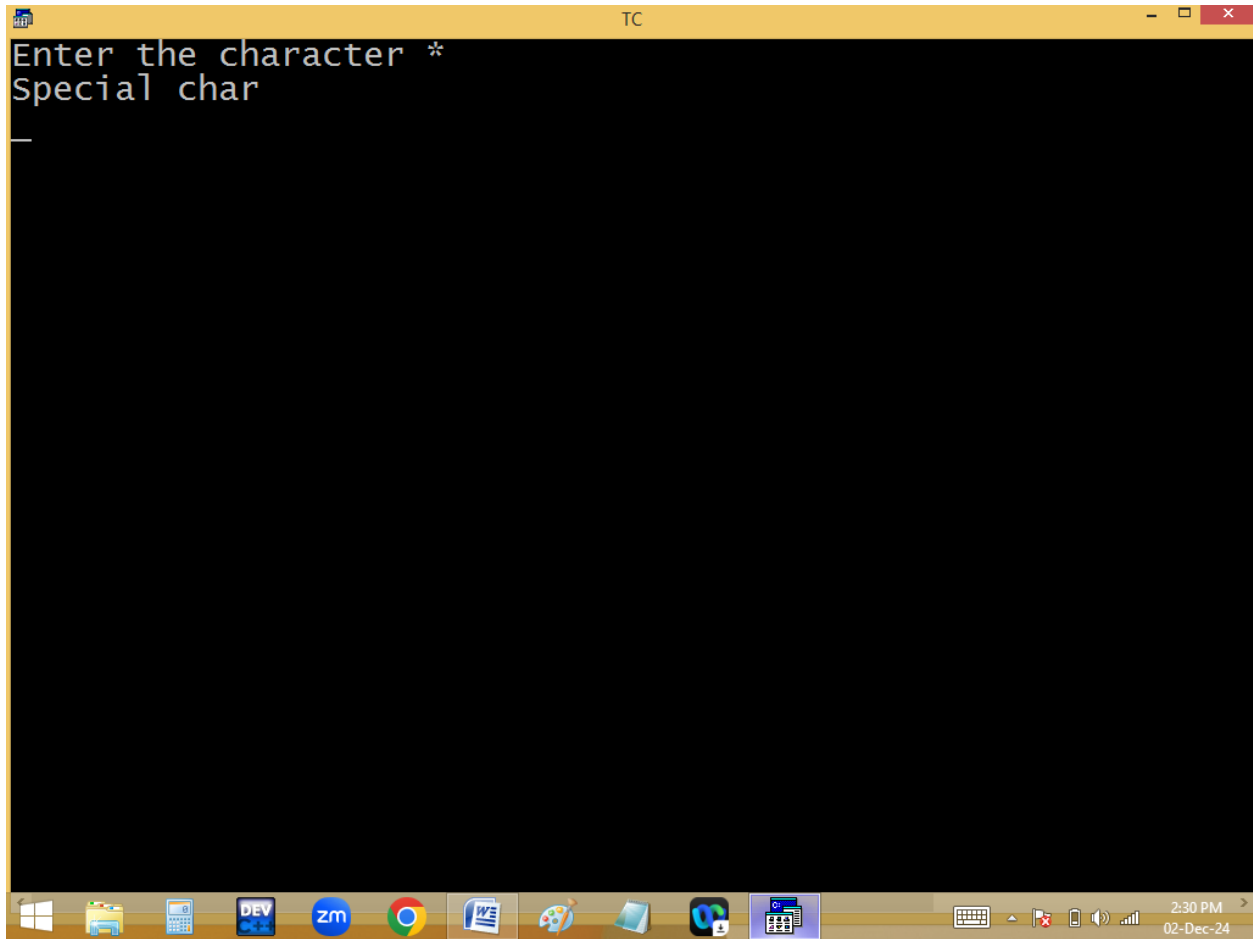
2:30 PM 02-Dec-24

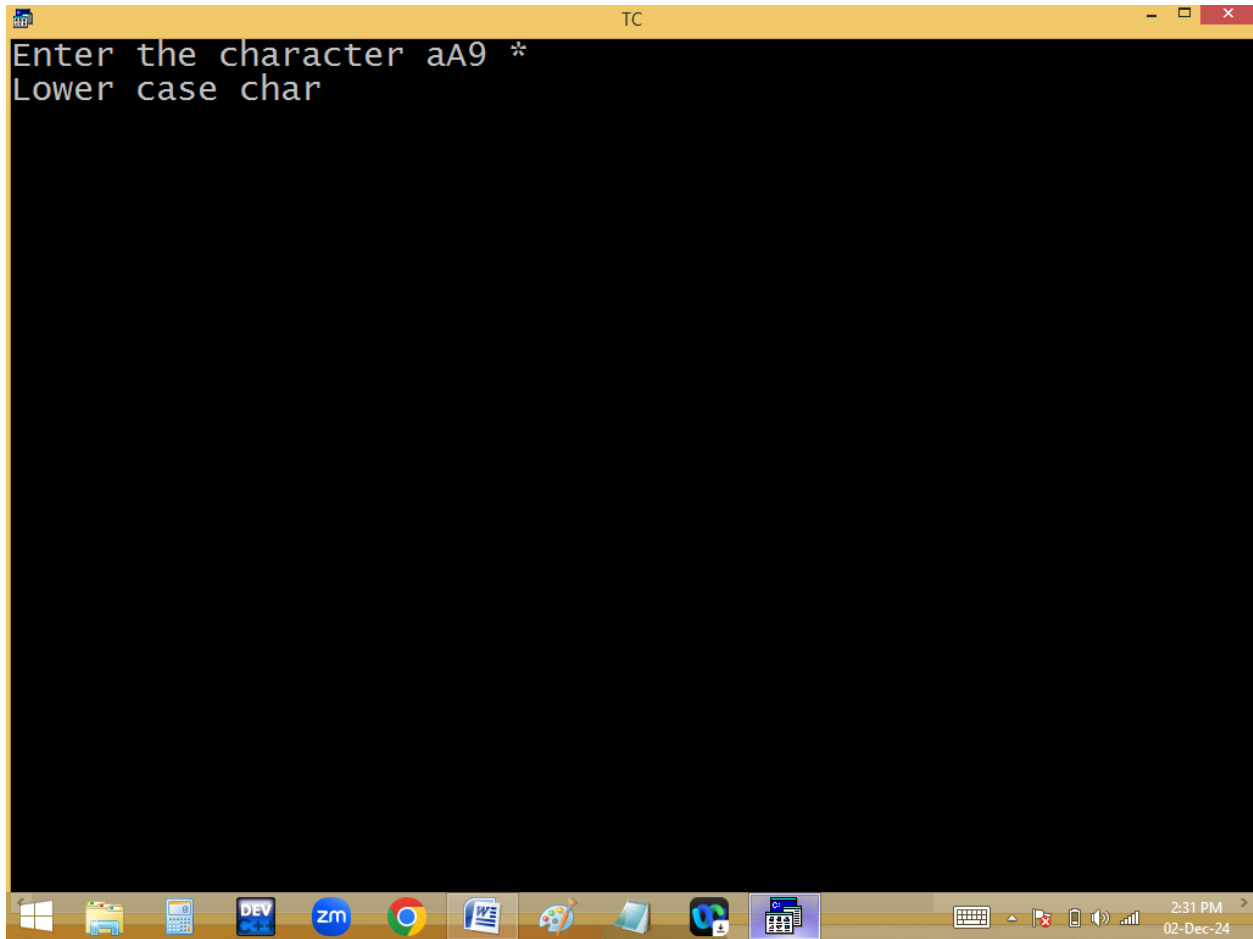


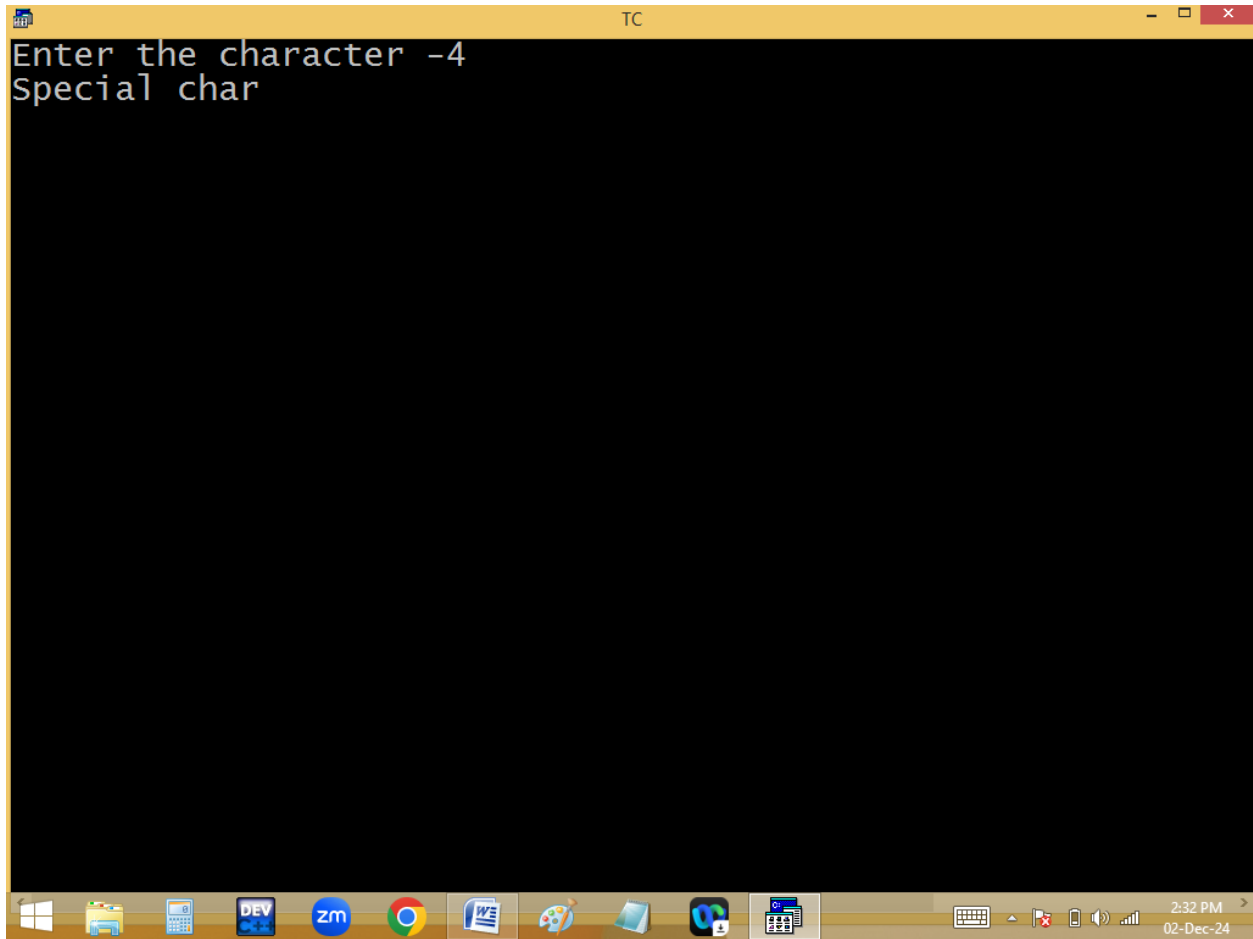




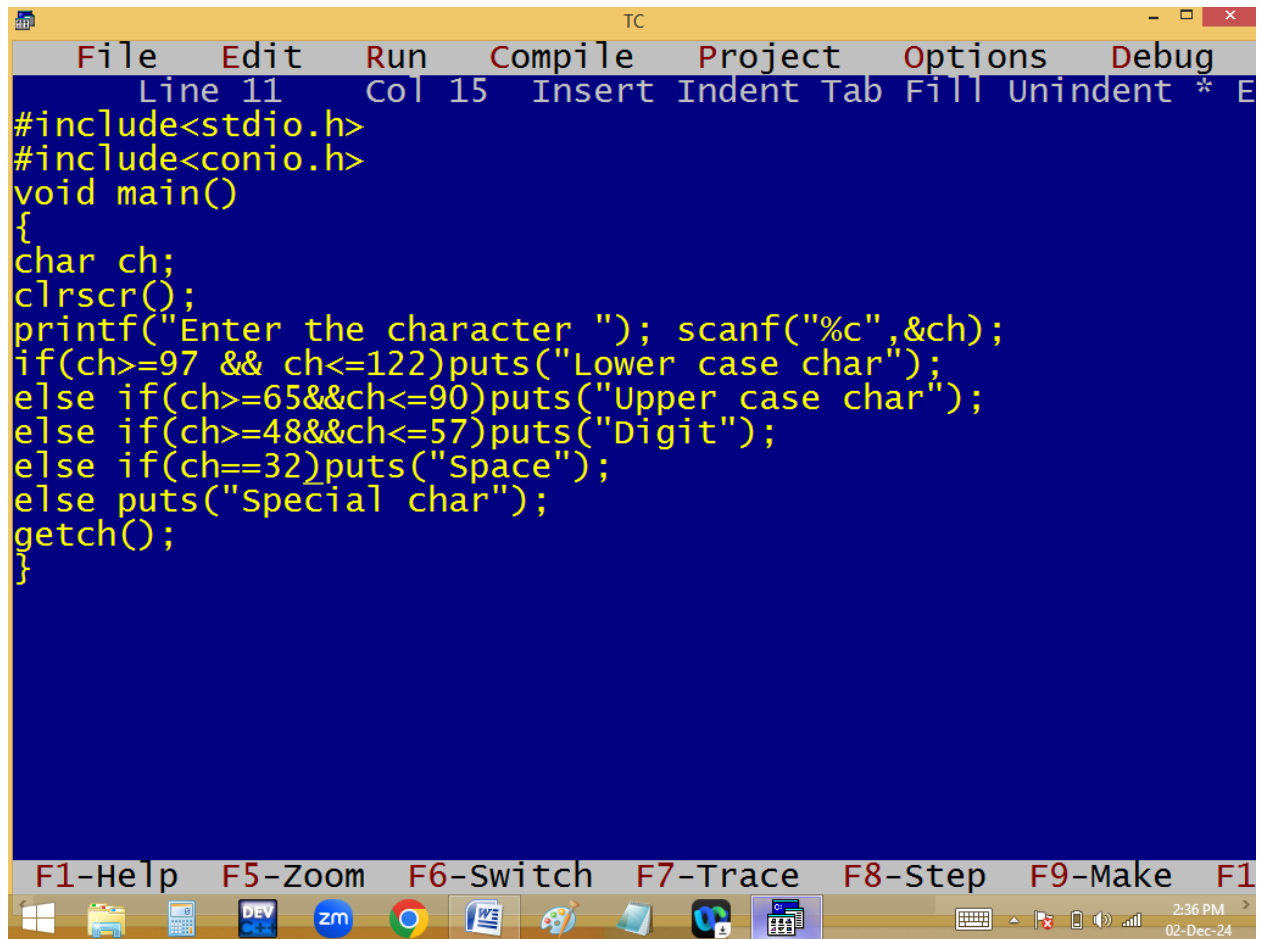








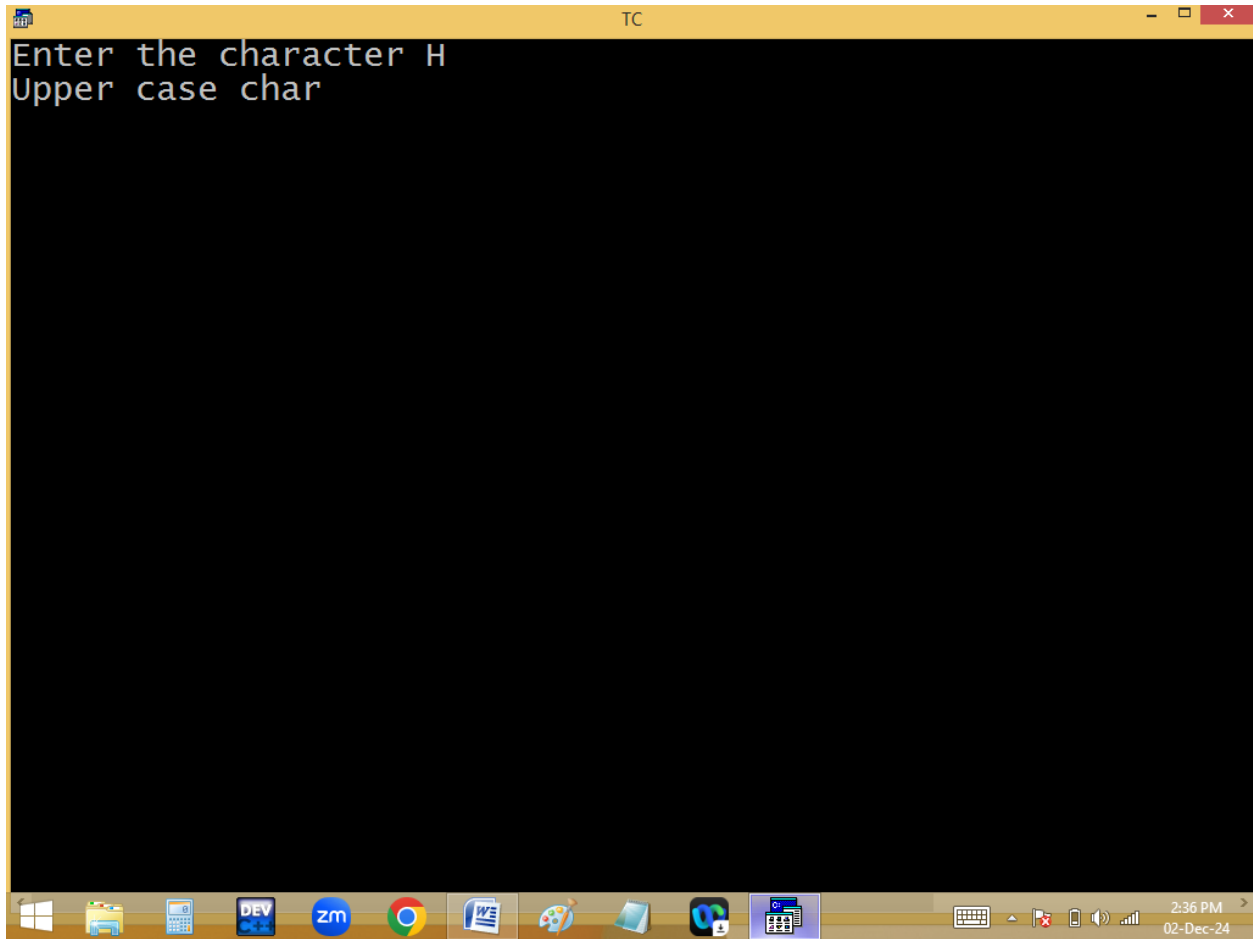
Using ascii values:

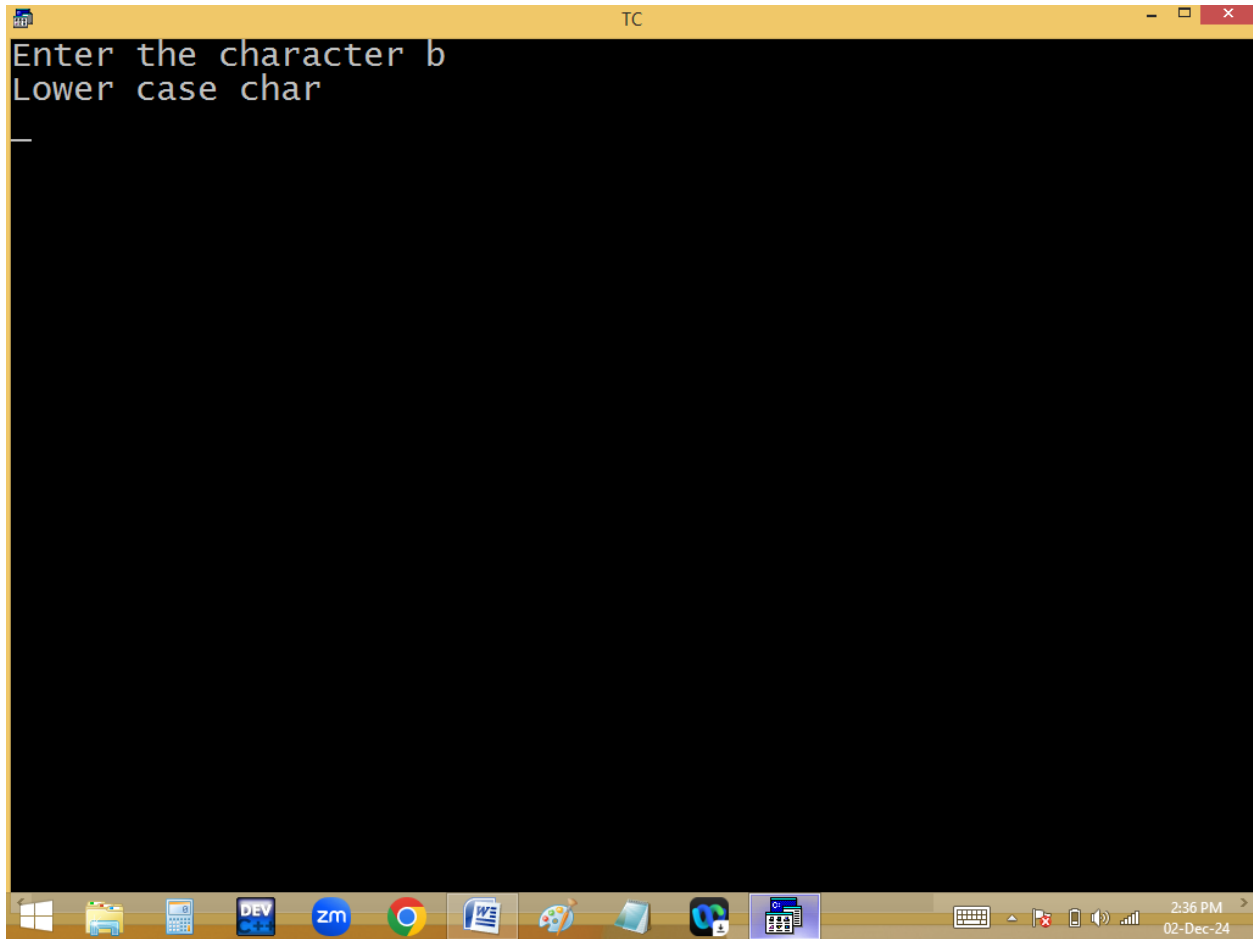


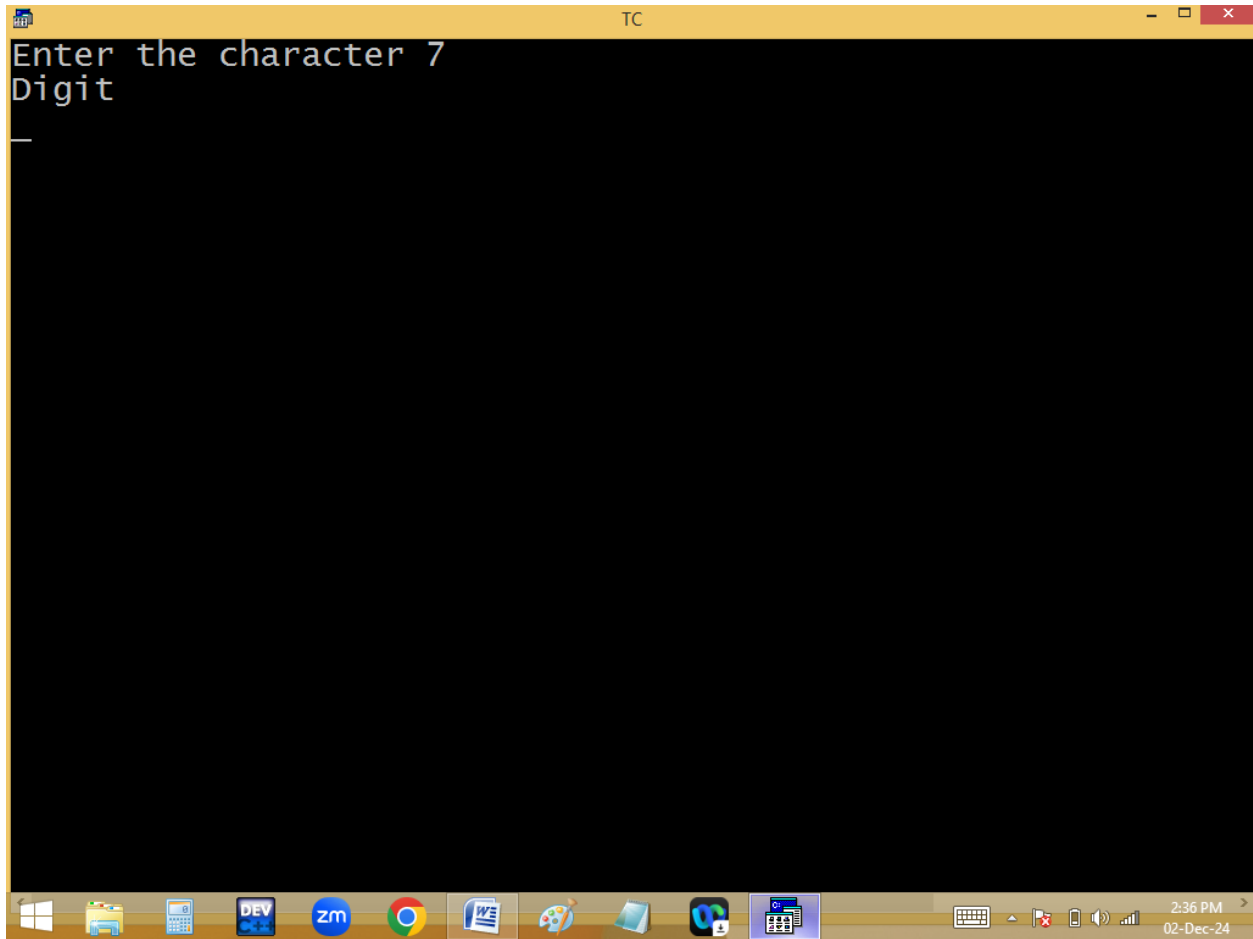
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 11", "Col 15", and "Insert Indent Tab Fill Unindent * E". The main editing area has a blue background and contains the following C code:

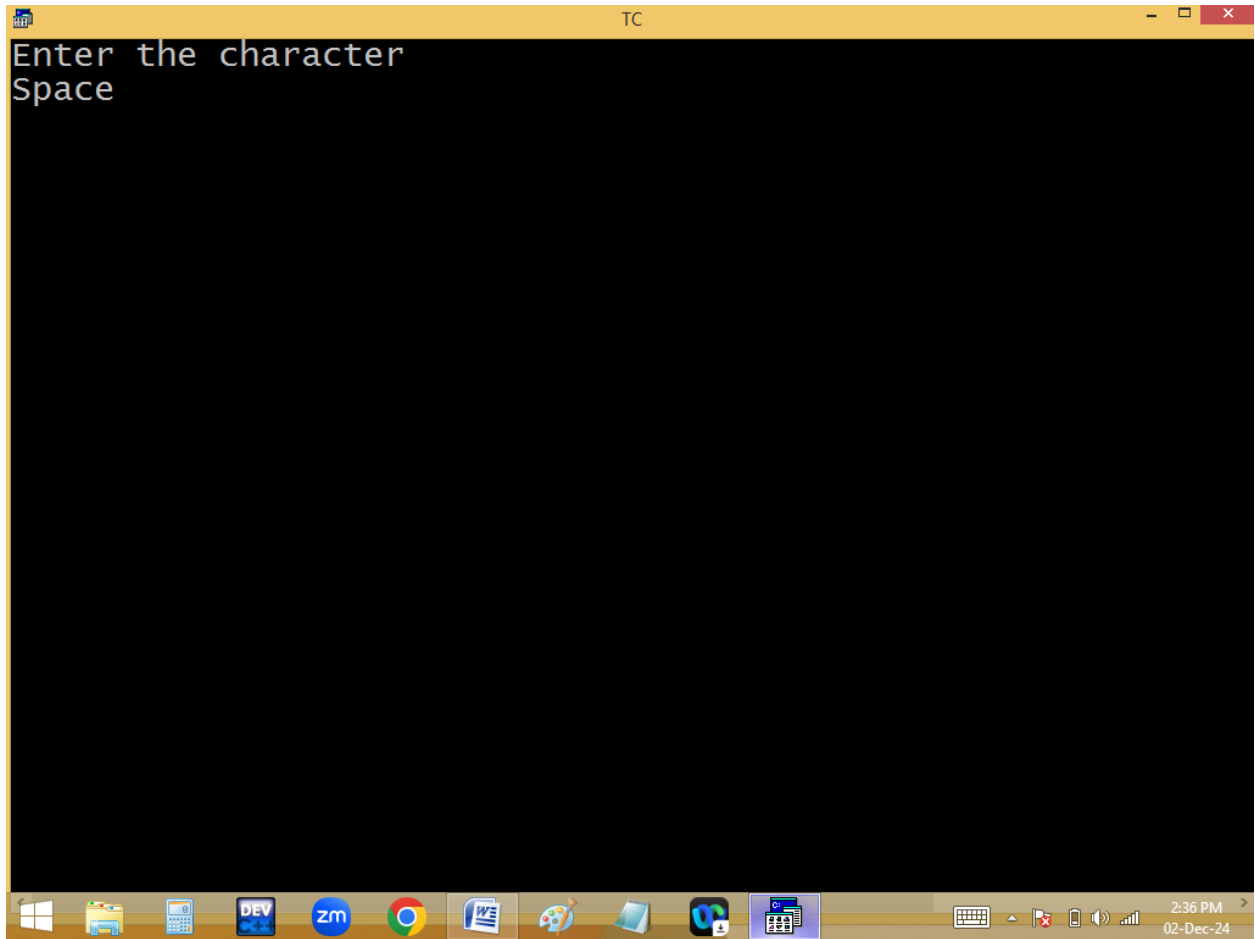
```
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(ch>=97 && ch<=122)puts("Lower case char");
else if(ch>=65&&ch<=90)puts("Upper case char");
else if(ch>=48&&ch<=57)puts("Digit");
else if(ch==32)puts("Space");
else puts("Special char");
getch();
}
```

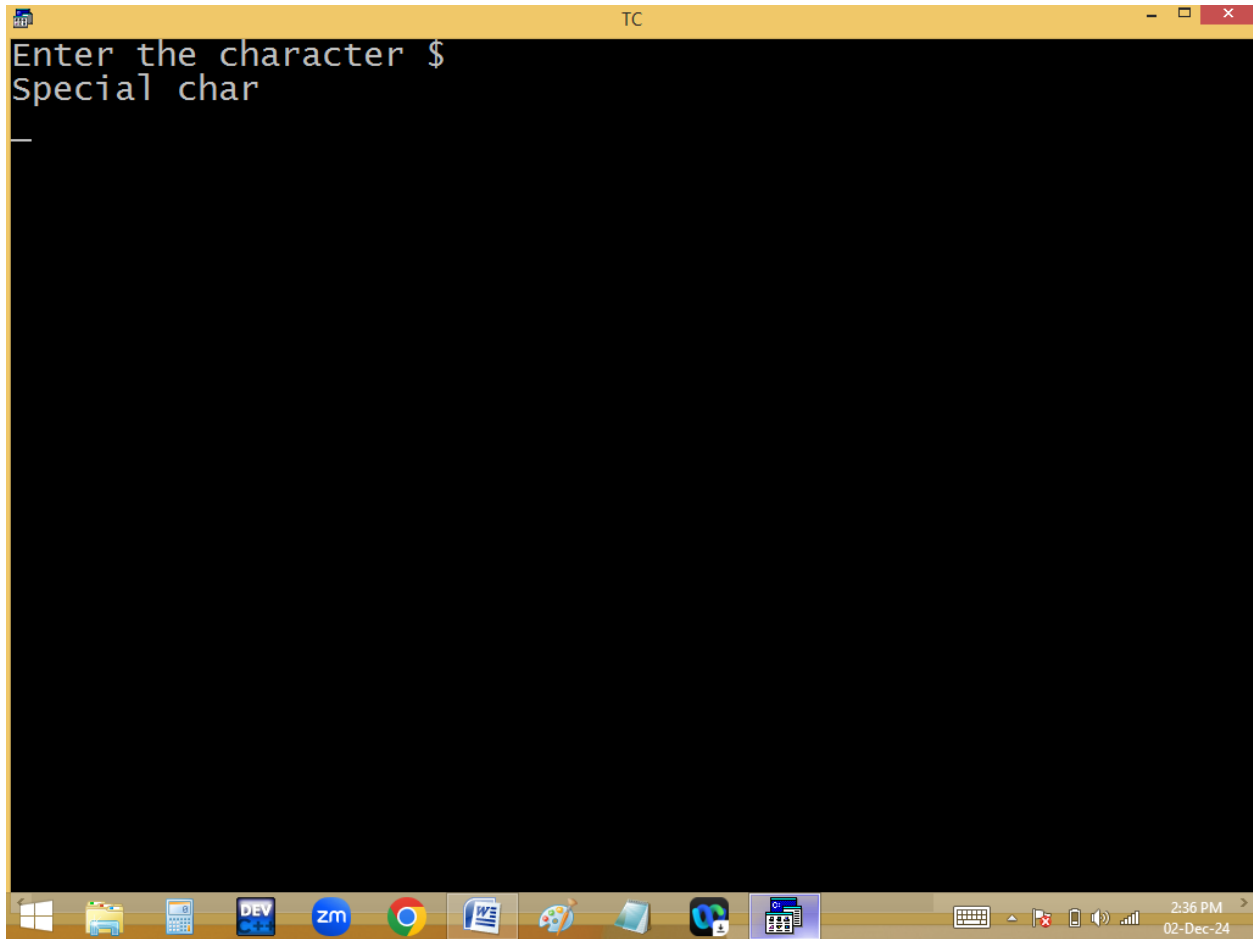
At the bottom of the IDE, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". Below the toolbar is the Windows taskbar, which includes icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Notepad, Paint, and the Turbo C++ application. The system tray on the right shows the time as "2:36 PM" and the date as "02-Dec-24".



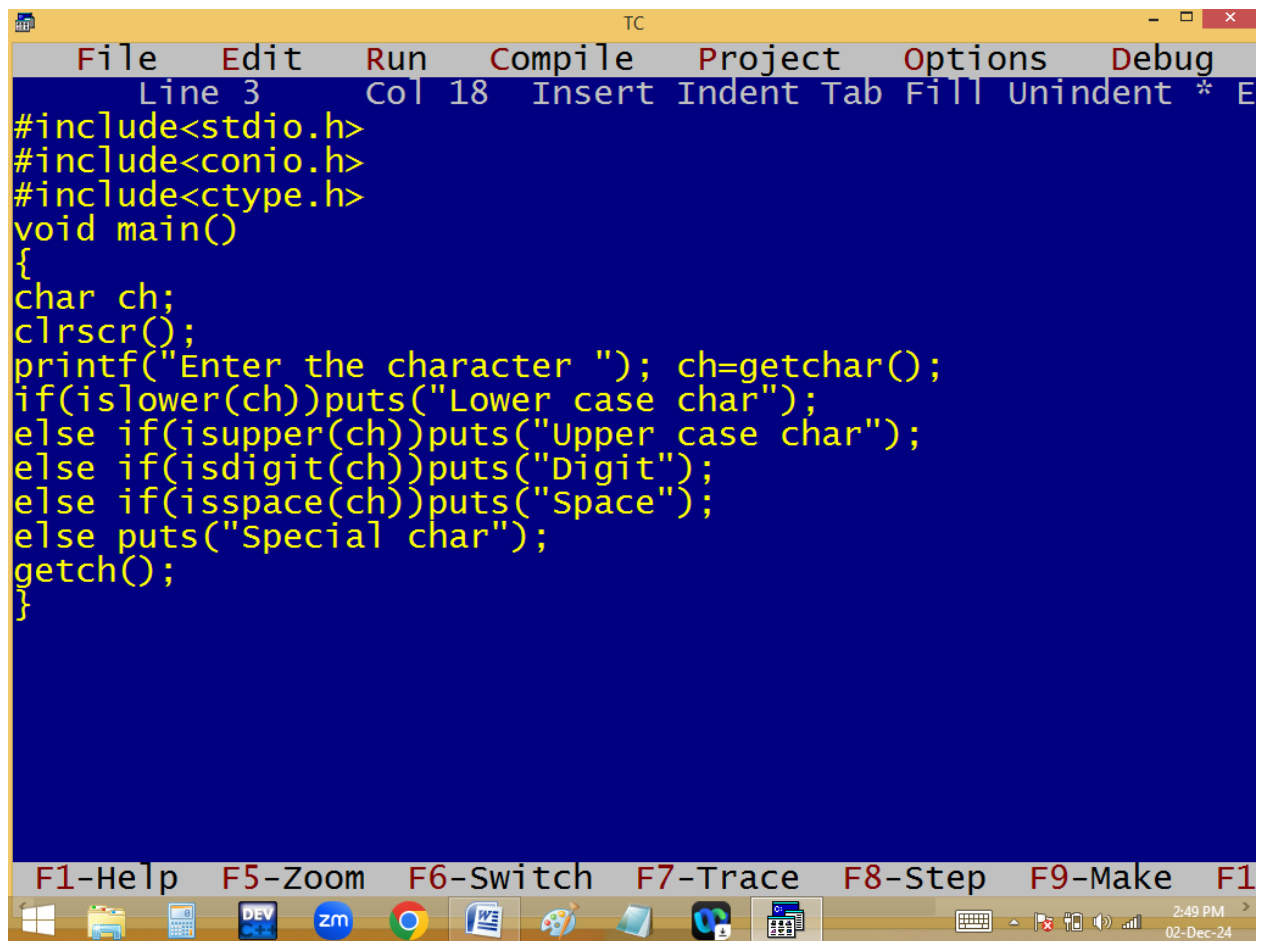








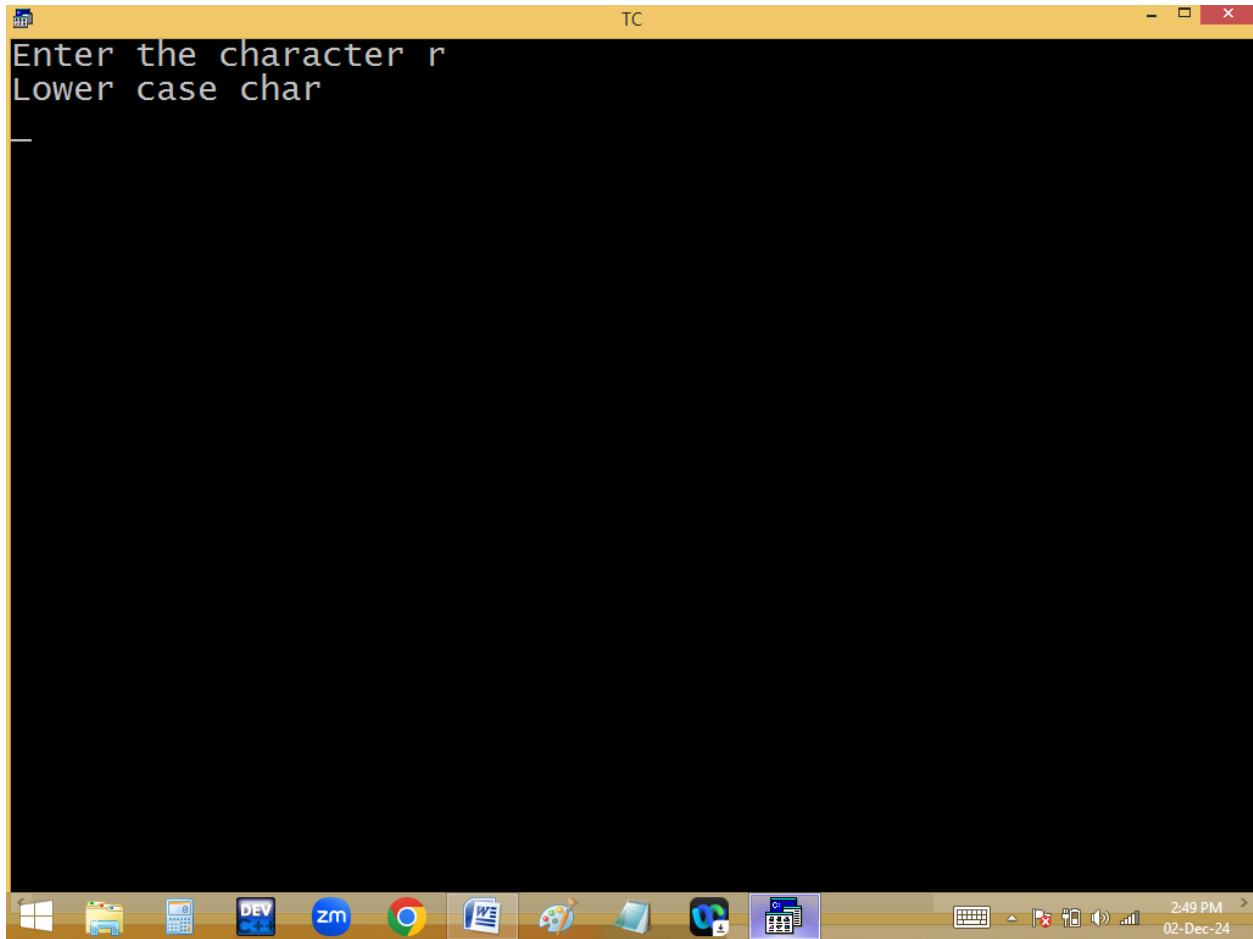
Using predefined functions:

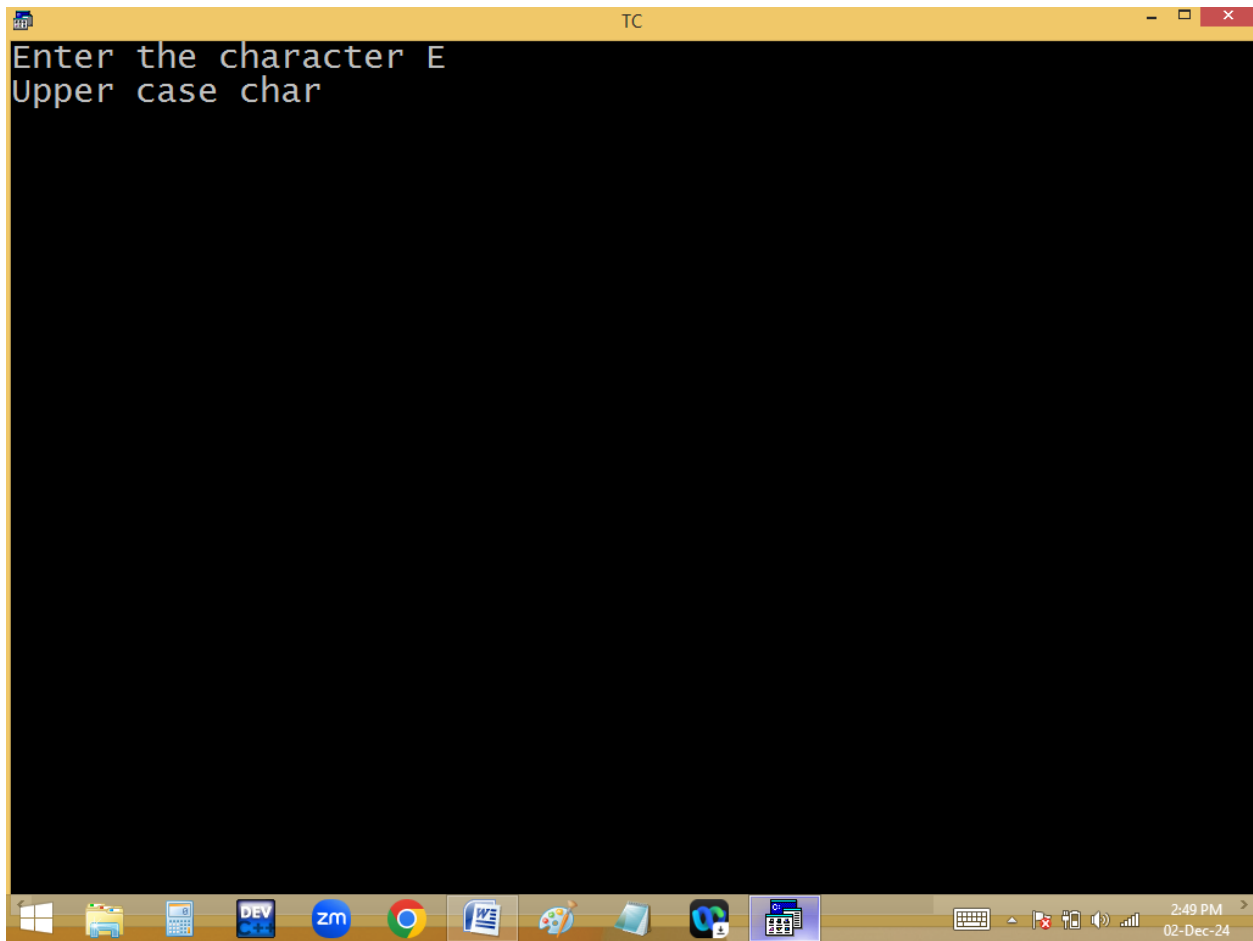


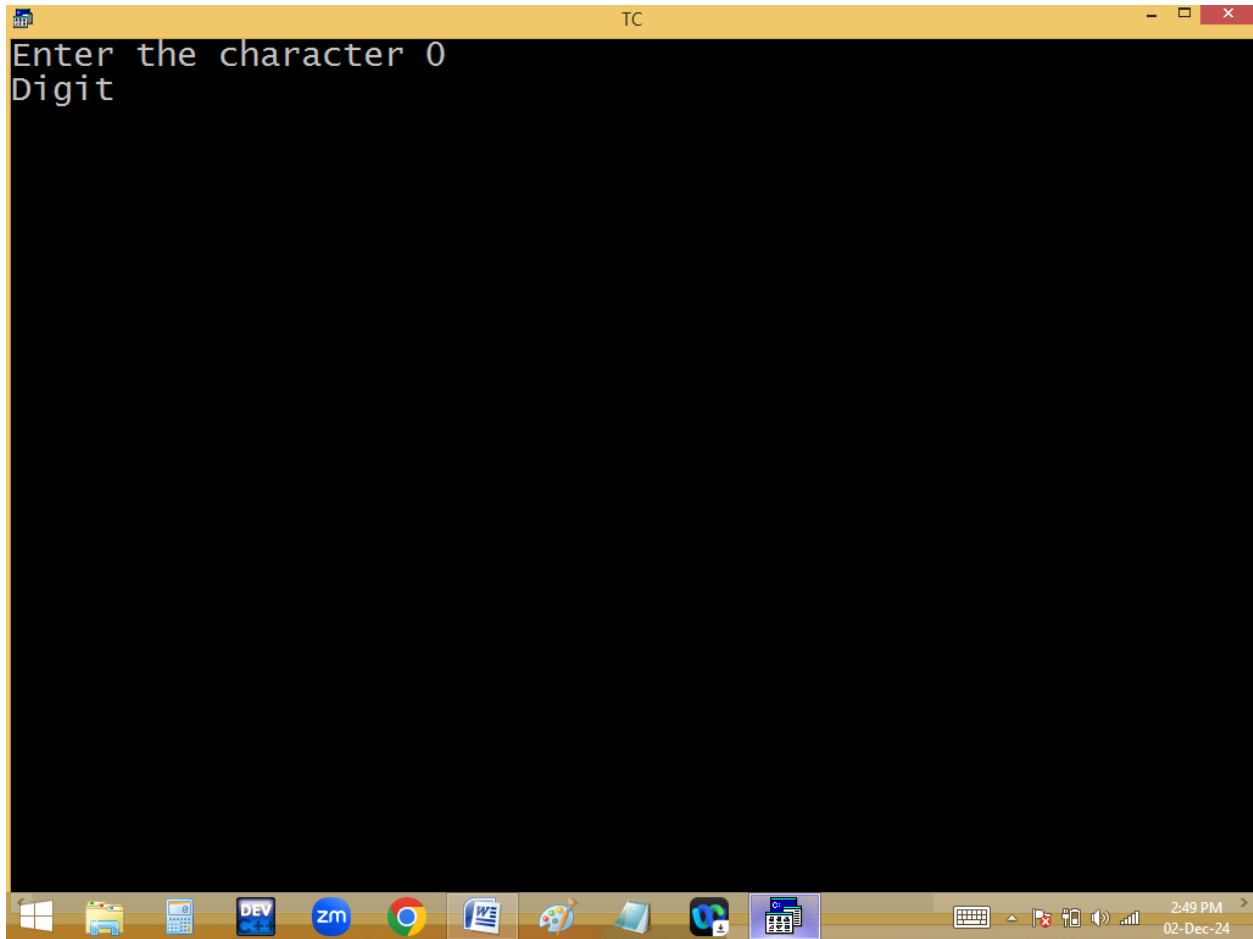
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 3", "Col 18", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a blue background and contains the following C code:

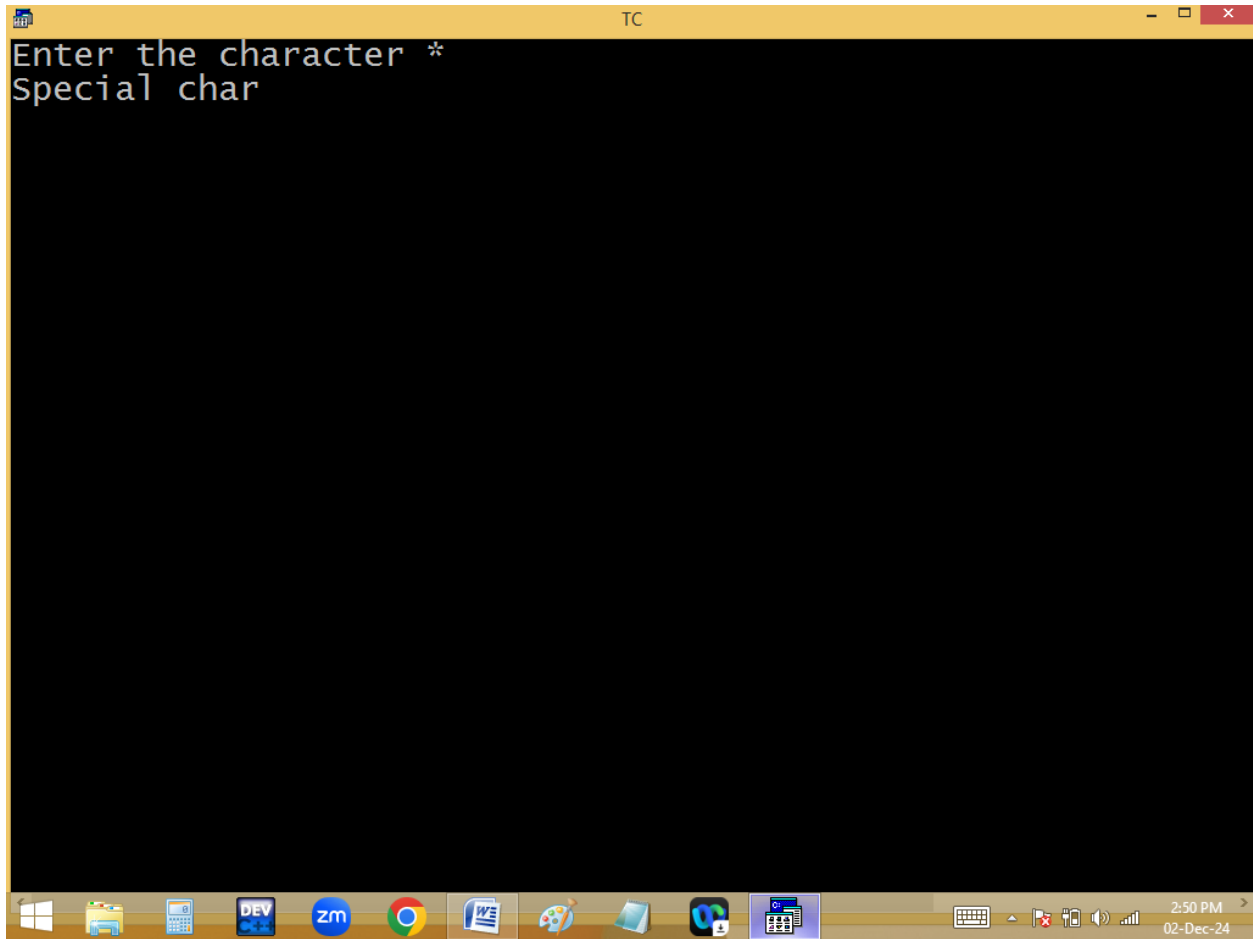
```
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); ch=getchar();
if(islower(ch))puts("Lower case char");
else if(isupper(ch))puts("Upper case char");
else if(isdigit(ch))puts("Digit");
else if(isspace(ch))puts("Space");
else puts("Special char");
getch();
}
```

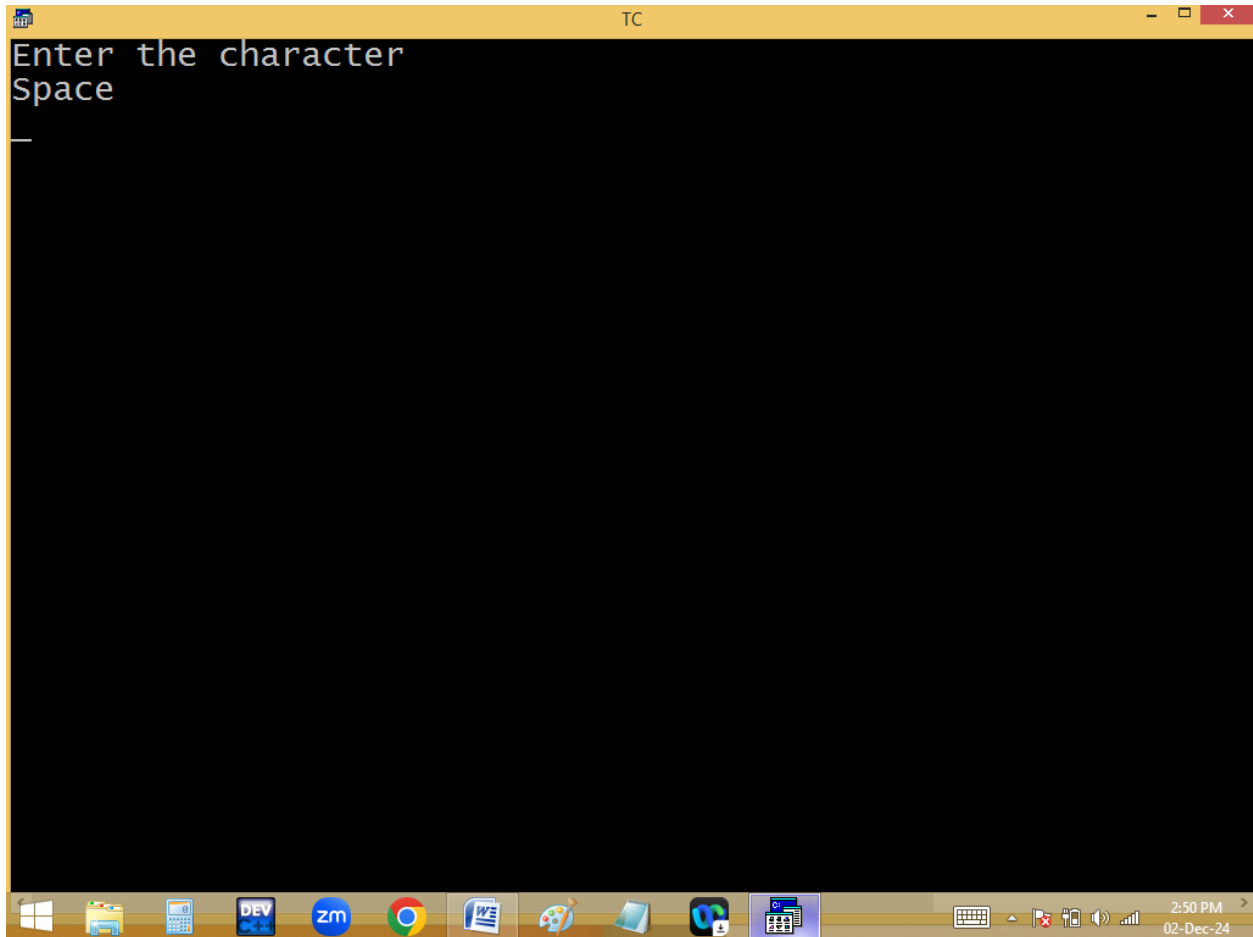
At the bottom of the window, there is a toolbar with icons for various functions. Below the toolbar, a row of function key shortcuts is displayed: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Exit". The Windows taskbar is visible at the very bottom, showing several application icons and the system clock indicating "2:49 PM" on "02-Dec-24".











getch()	getche()	getchar()
Reads single char	Reads single char	Reads single char
Entered char is not visible	char is visible	char is visible
Not waiting for enter key	Not waiting for enter key	Waiting for enter key
Belongs to conio.h	Belongs to conio.h	Belongs to stdio.h
char not changeable	char not changeable	char changeable

Note: All the above functions returns ASCII value(int) of reading character.

isupper(), islower(), isdigit(), isspace():

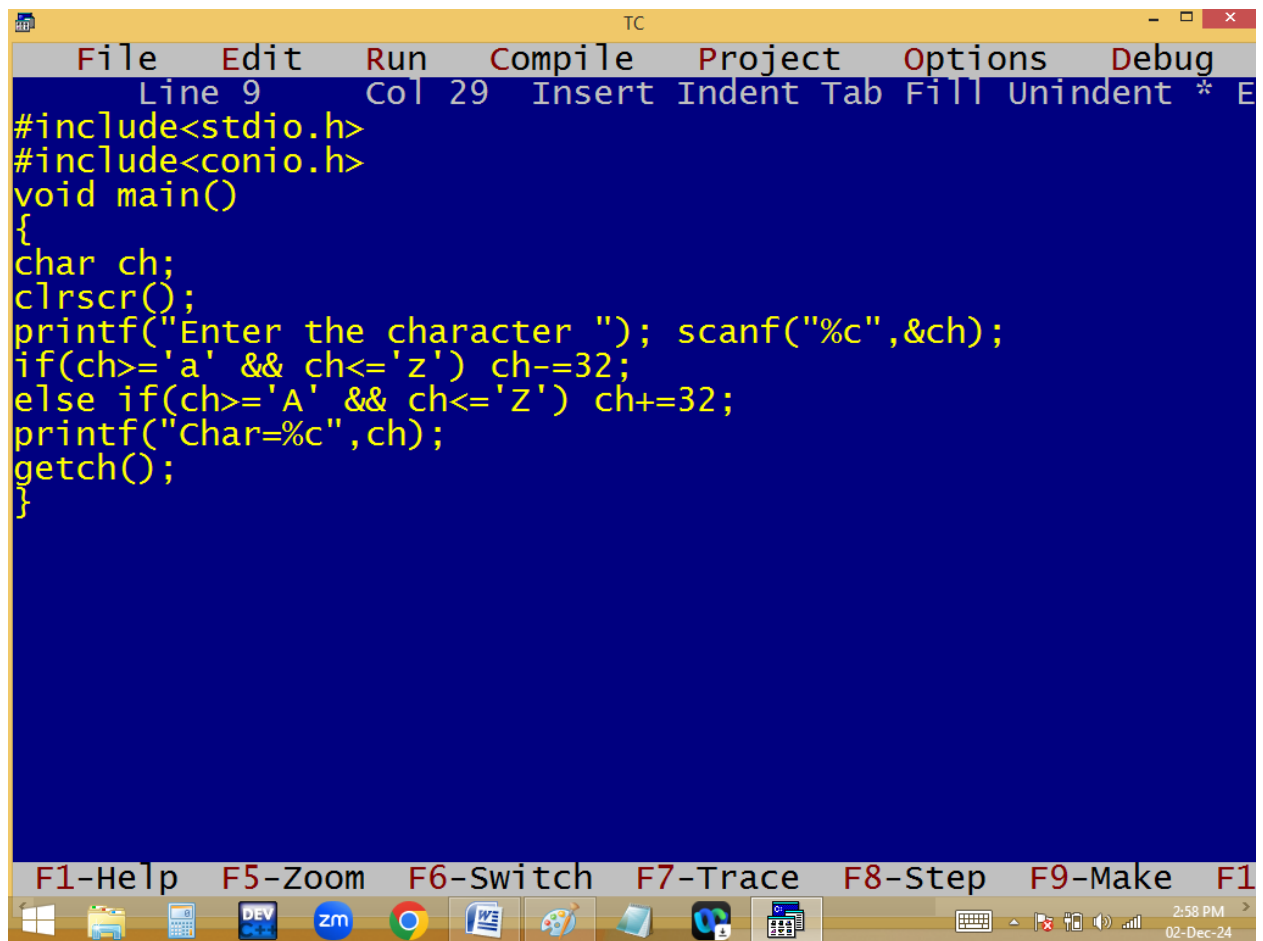
They are used to check the given character is upper / lower / digit / space.

tolower(), toupper(): They are used to convert the character from upper case to lower and lower to upper.

All these functions / macros available in `<ctype.h>`

Lower to upper / upper to lower conversion:

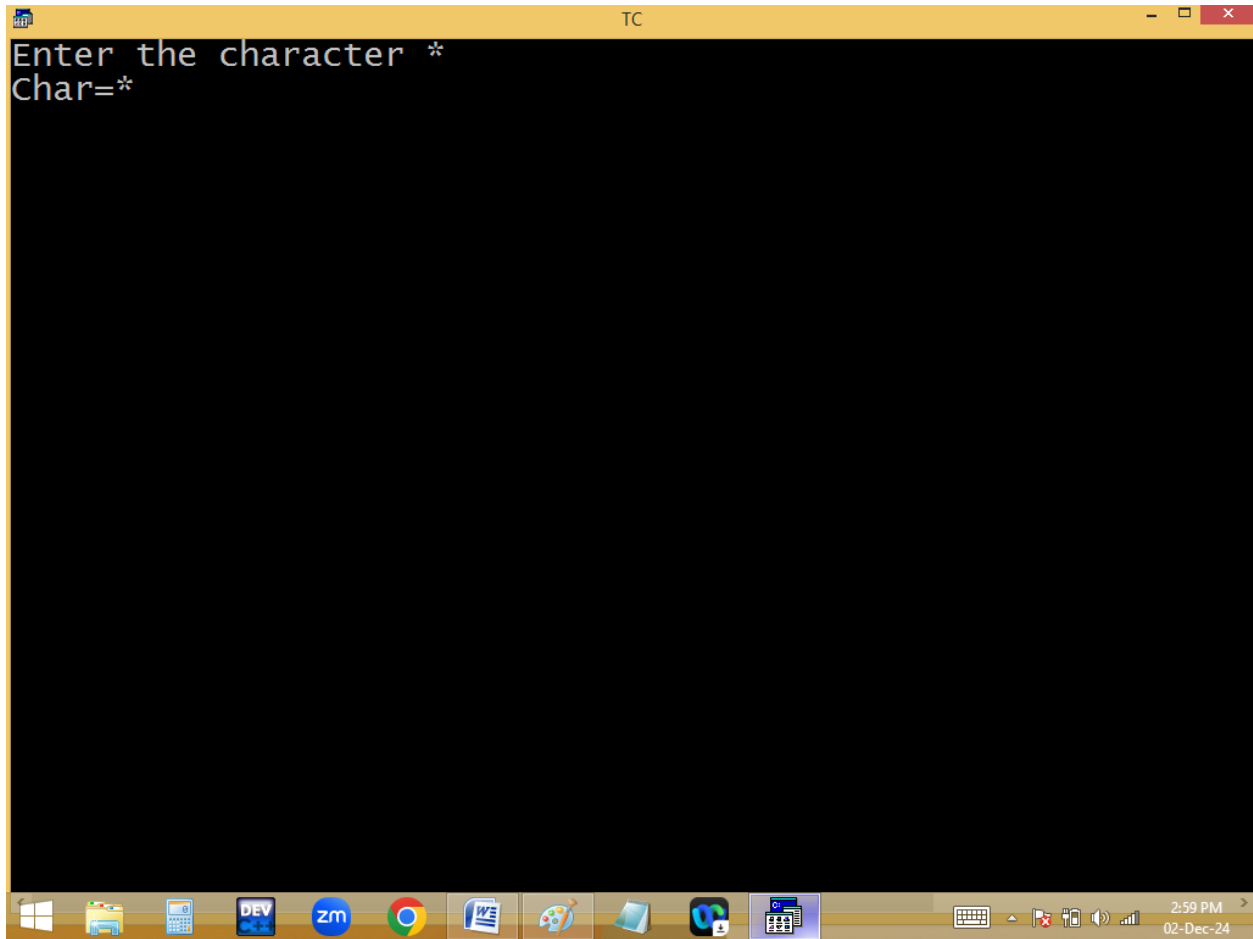
$$\begin{array}{r} a = 97 \\ - 32 \\ \hline A = 65 \\ + 32 \\ \hline a = 97 \end{array}$$

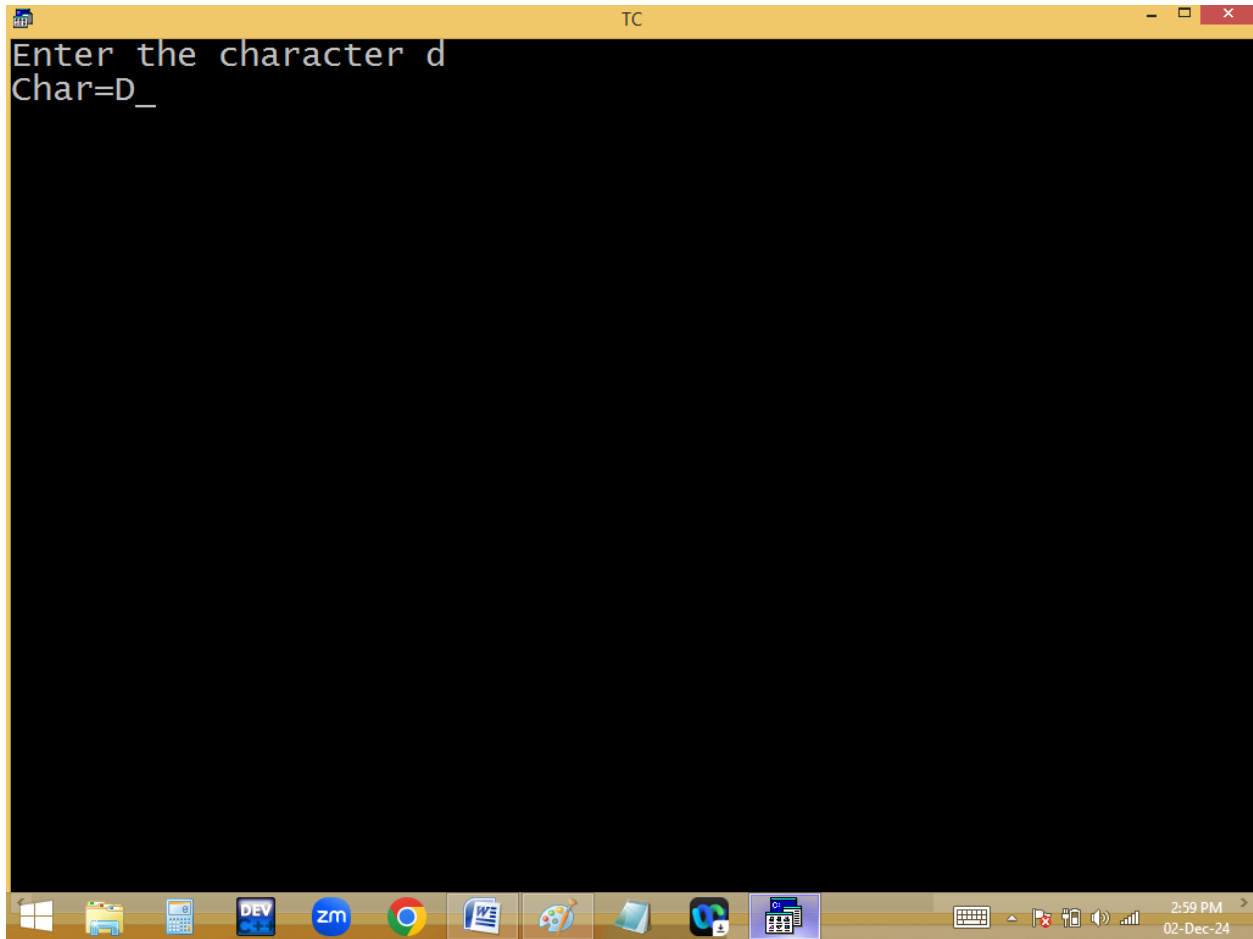


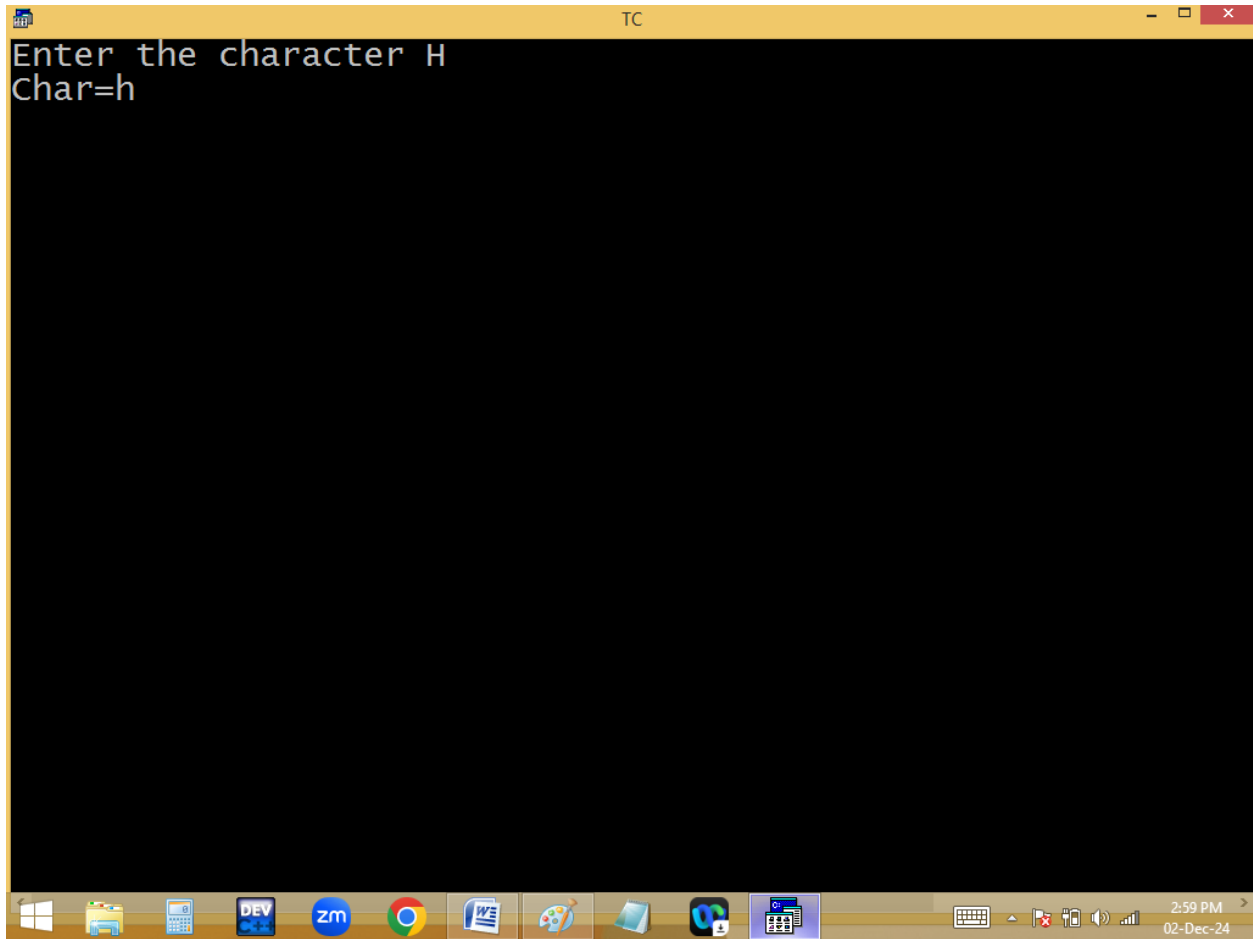
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 9", "Col 29", and lists editing options: "Insert", "Indent", "Tab", "Fill", "Unindent", and a "E" (likely for End). The main editing area has a blue background and contains the following C code:

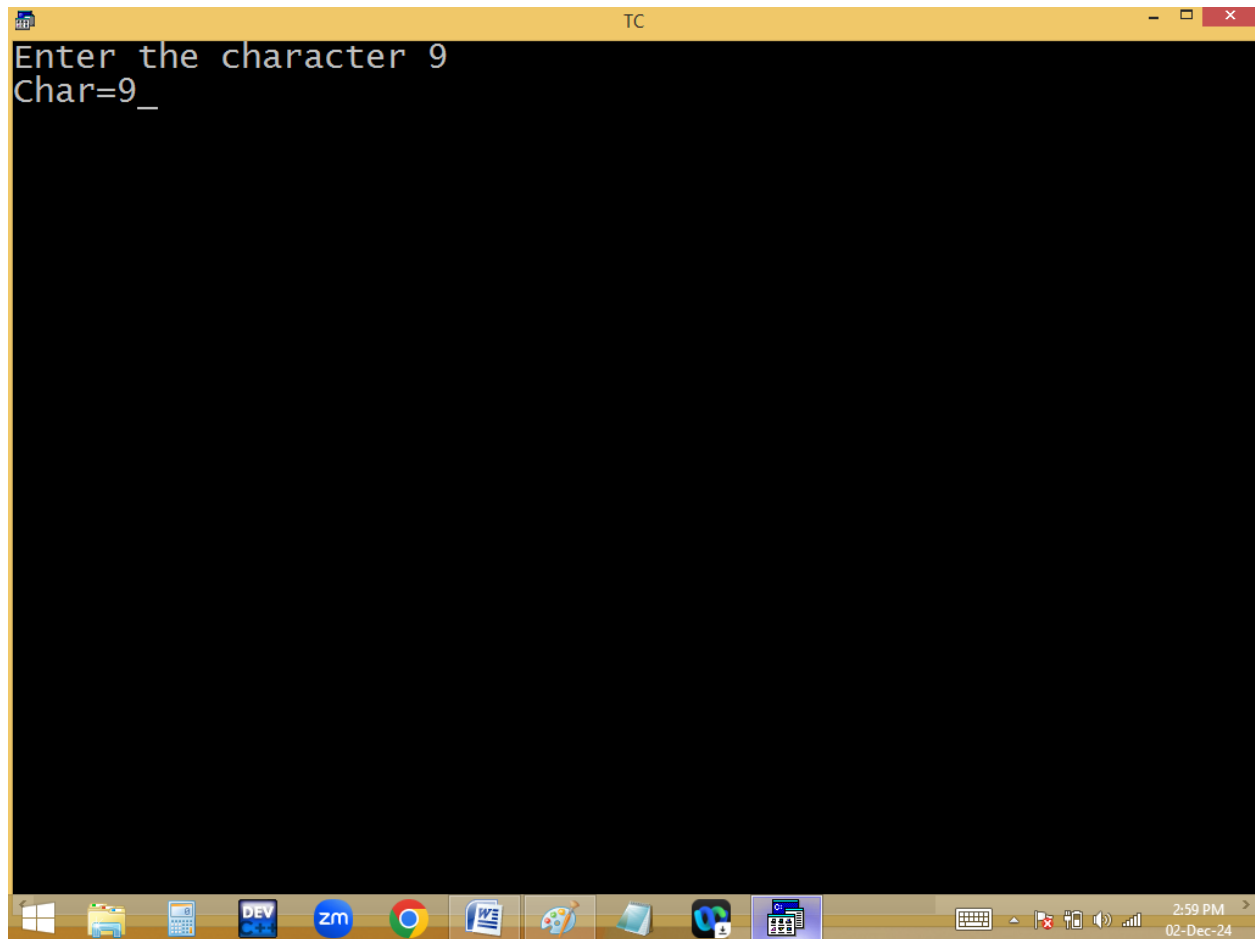
```
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(ch>='a' && ch<='z') ch-=32;
else if(ch>='A' && ch<='Z') ch+=32;
printf("Char=%c",ch);
getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and the Turbo C++ application. The system clock in the bottom right corner shows "2:58 PM" and "02-Dec-24".

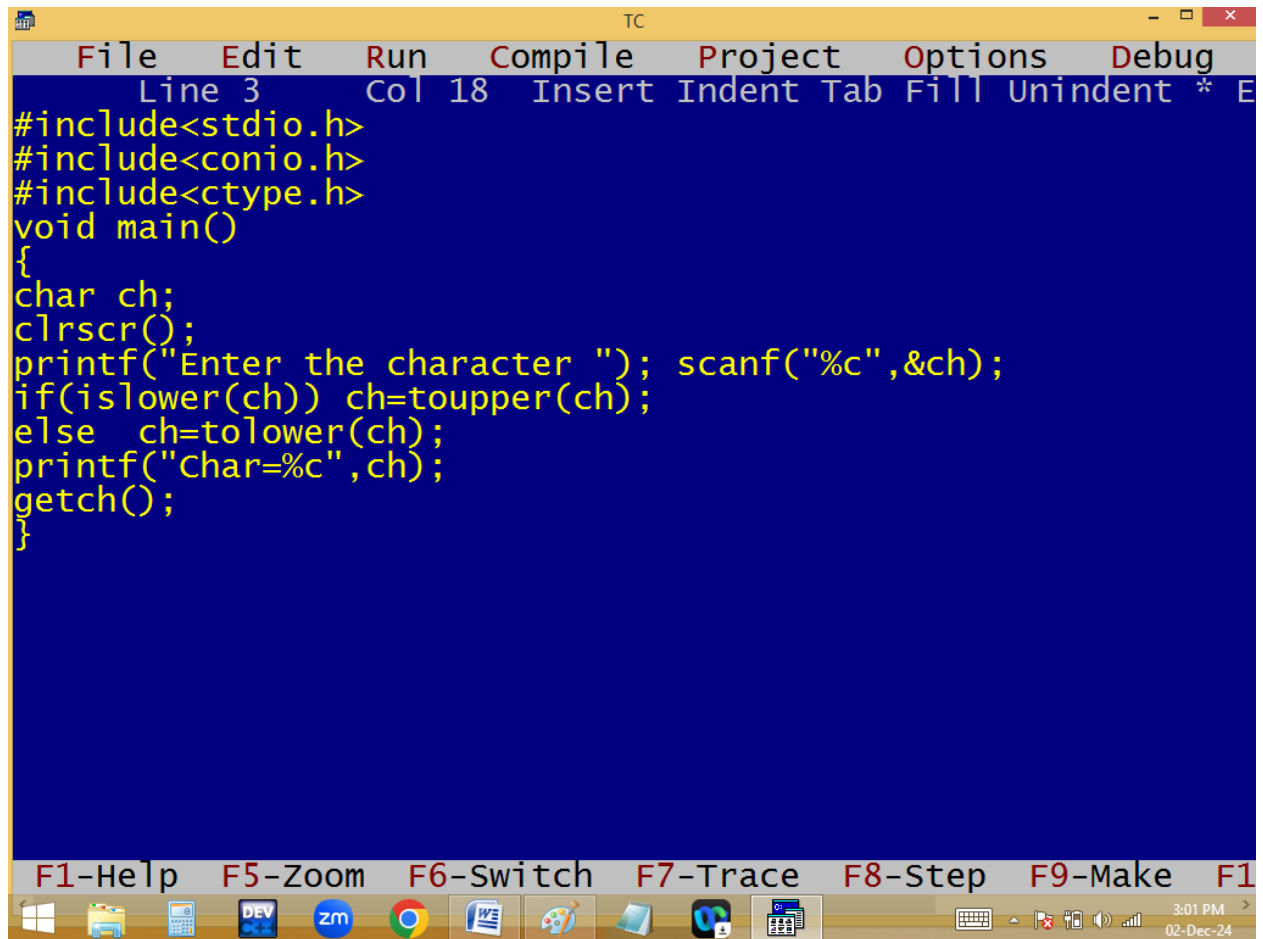








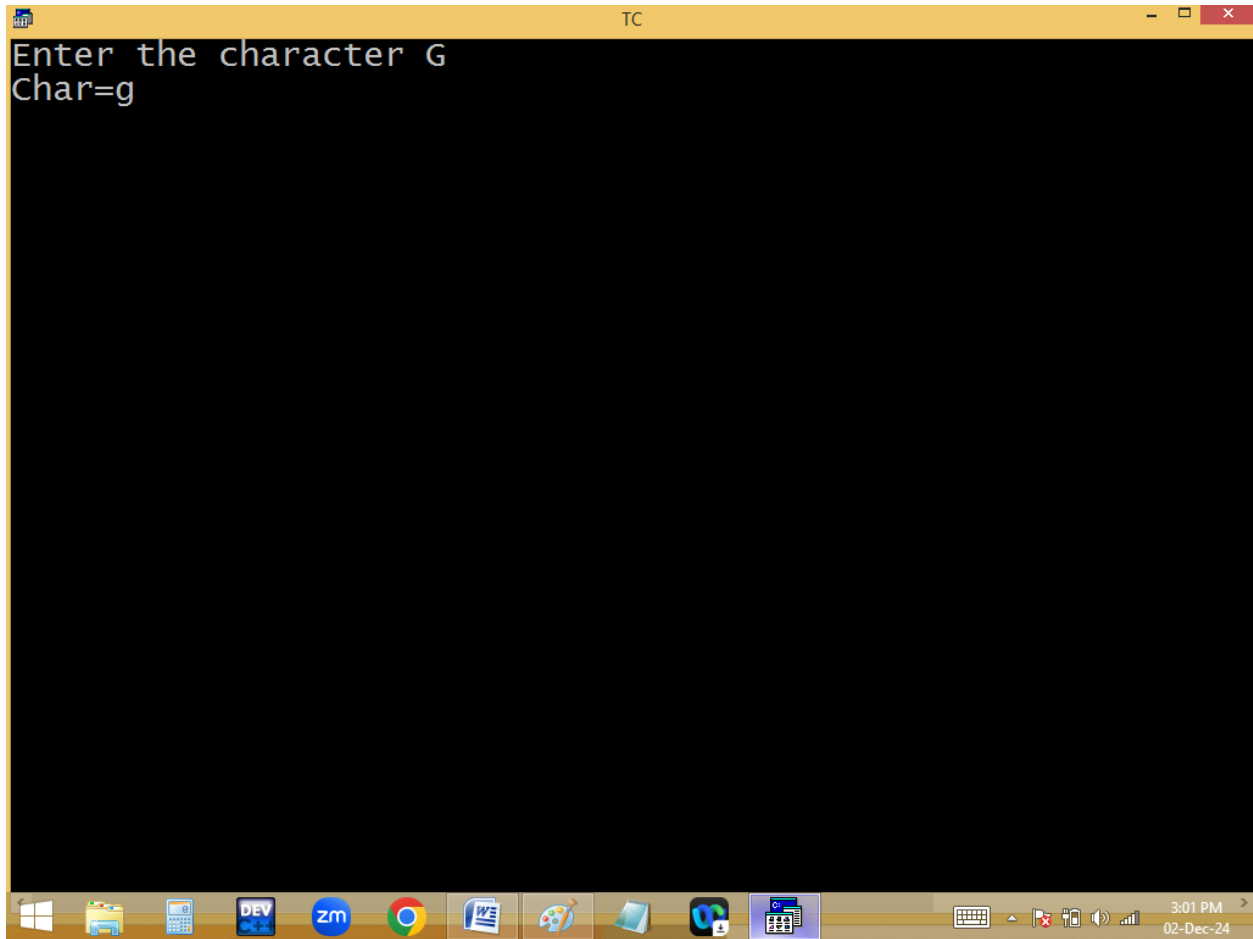
Using predefined functions:

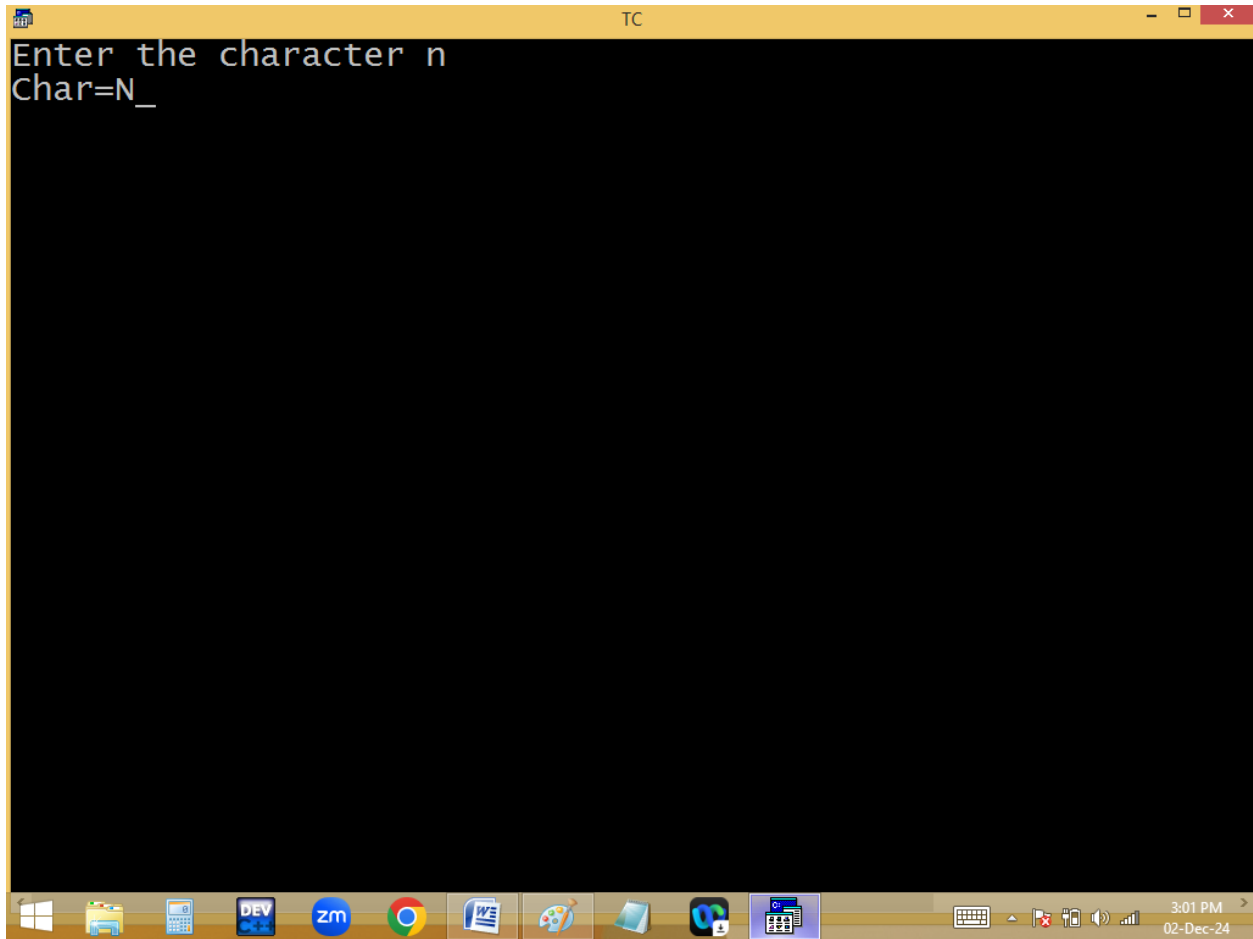


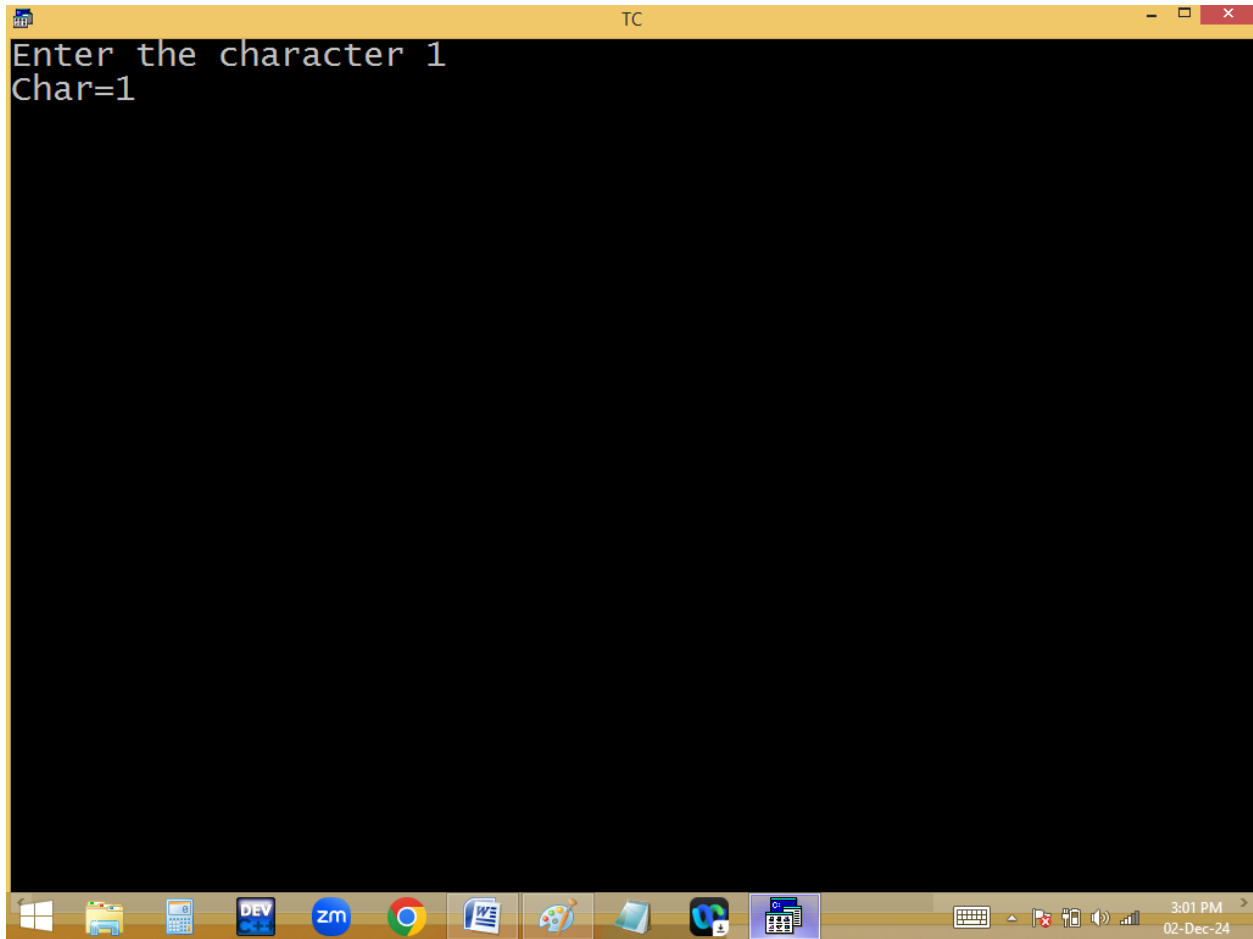
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 3", "Col 18", and various editing options like "Insert", "Indent", "Tab", "Fill", "Unindent", and a multiplier "* E". The main editing area has a blue background and contains the following C code:

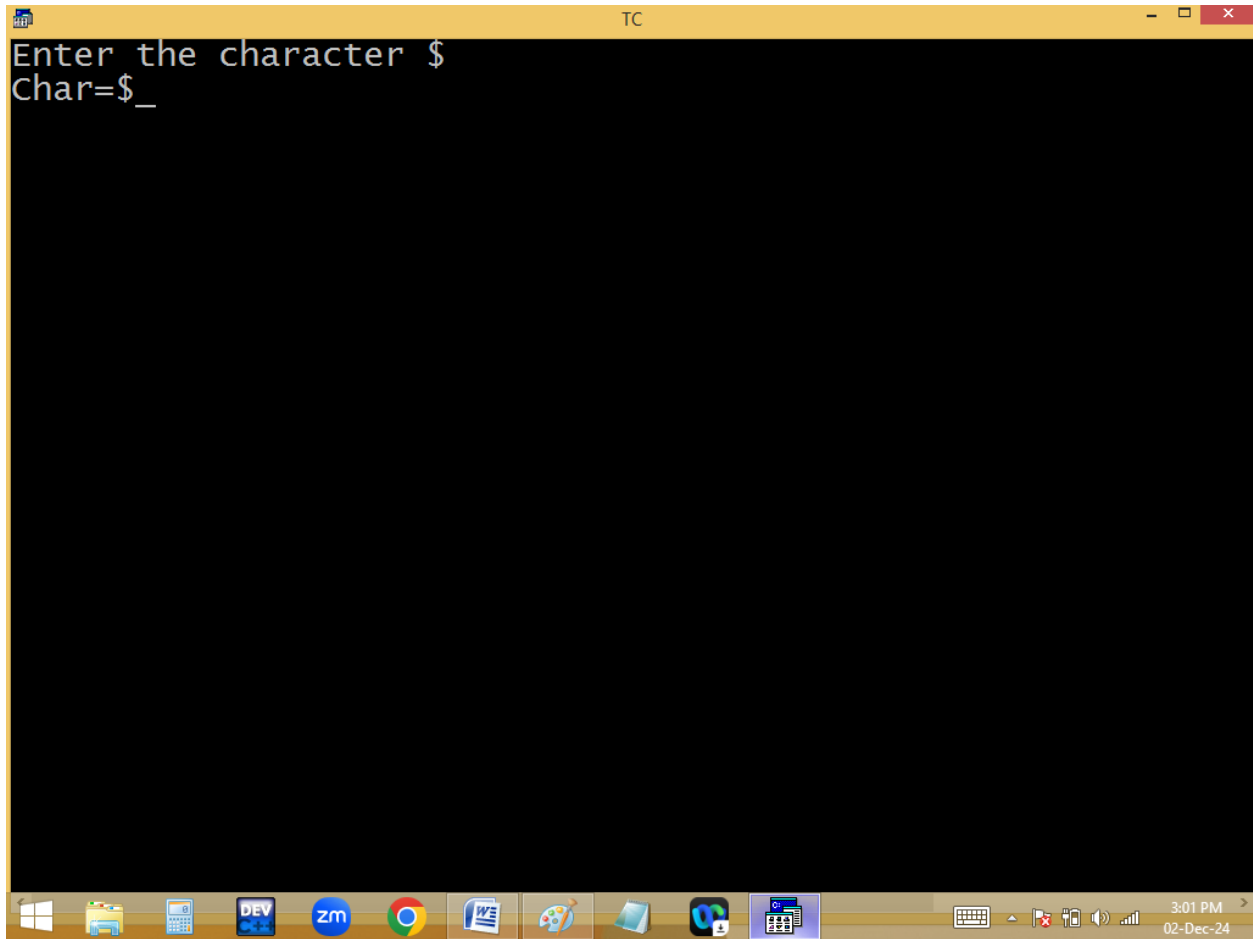
```
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(islower(ch)) ch=toupper(ch);
else ch=tolower(ch);
printf("Char=%c",ch);
getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions and a status bar. The status bar displays the time "3:01 PM" and the date "02-Dec-24".



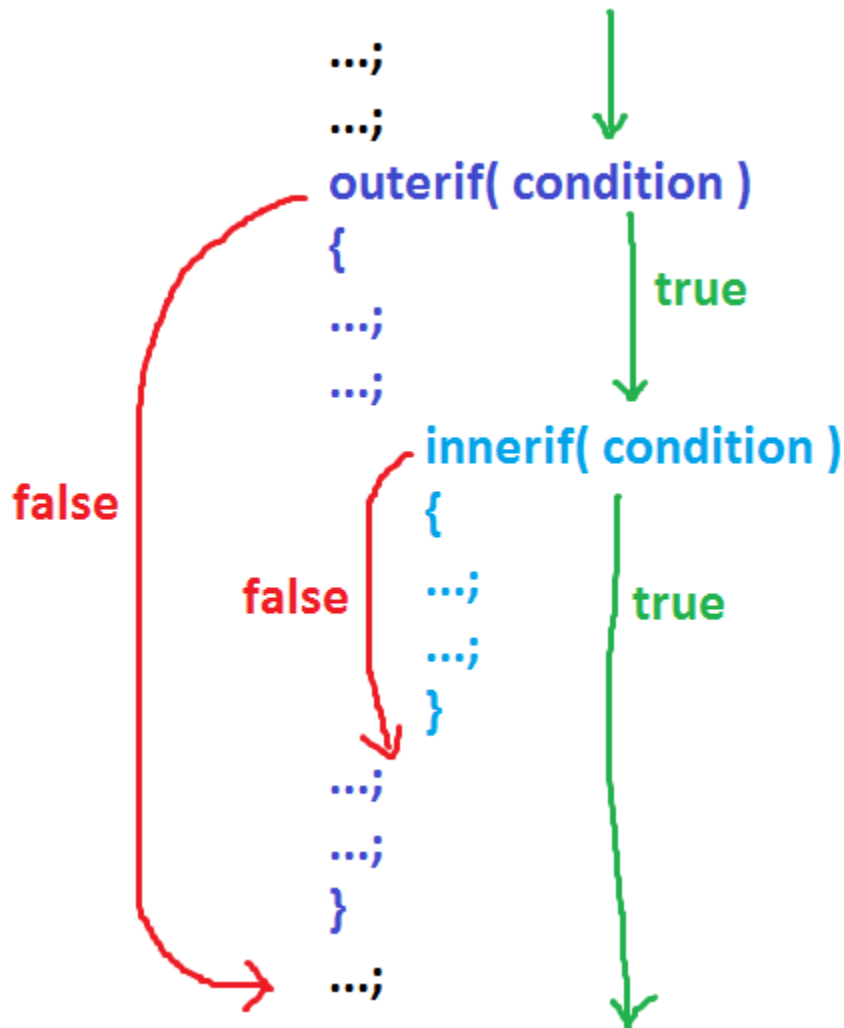






Nested if:

One if within another if called nested if.

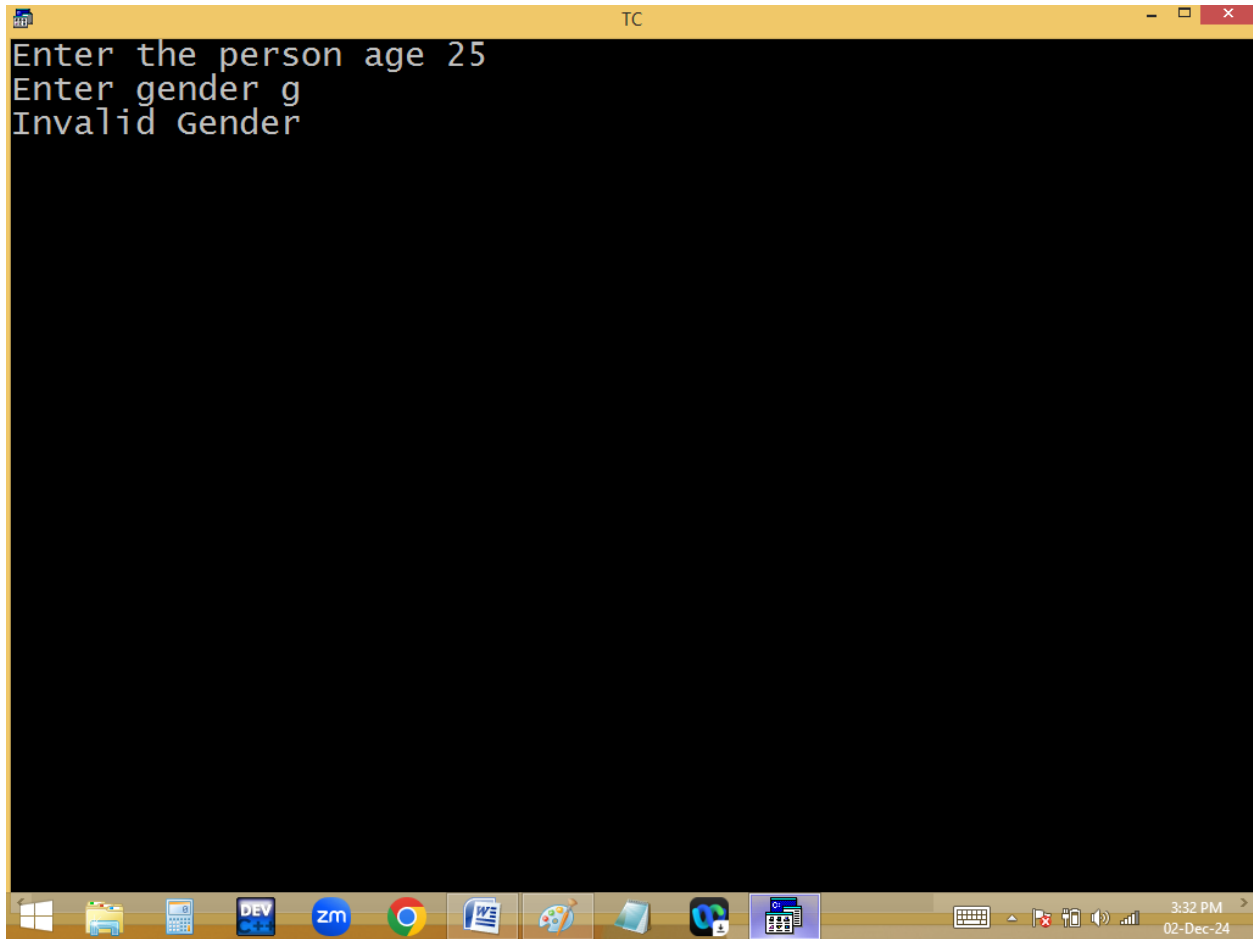


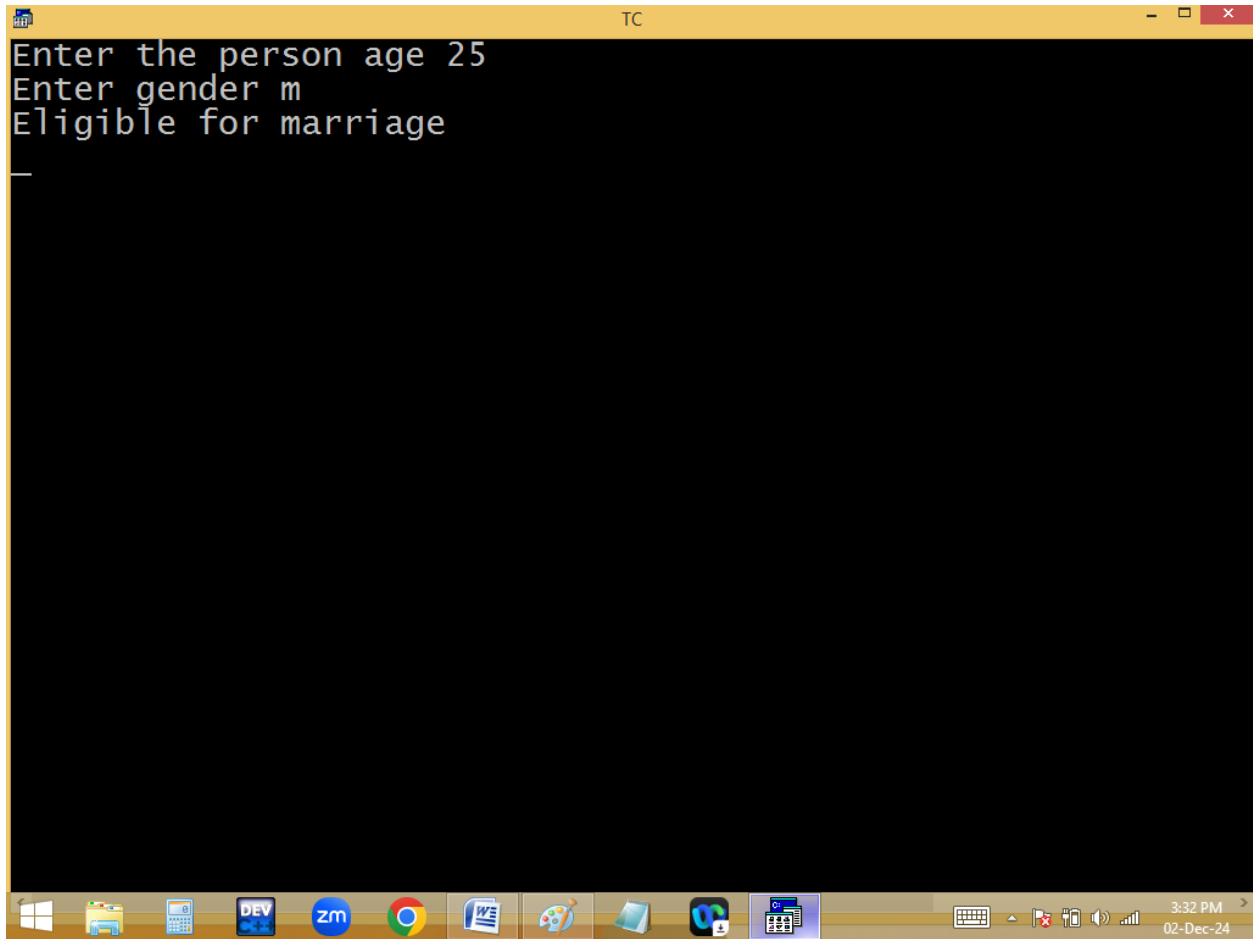
Read a person age and gender and determine that person is eligible for marriage or not?

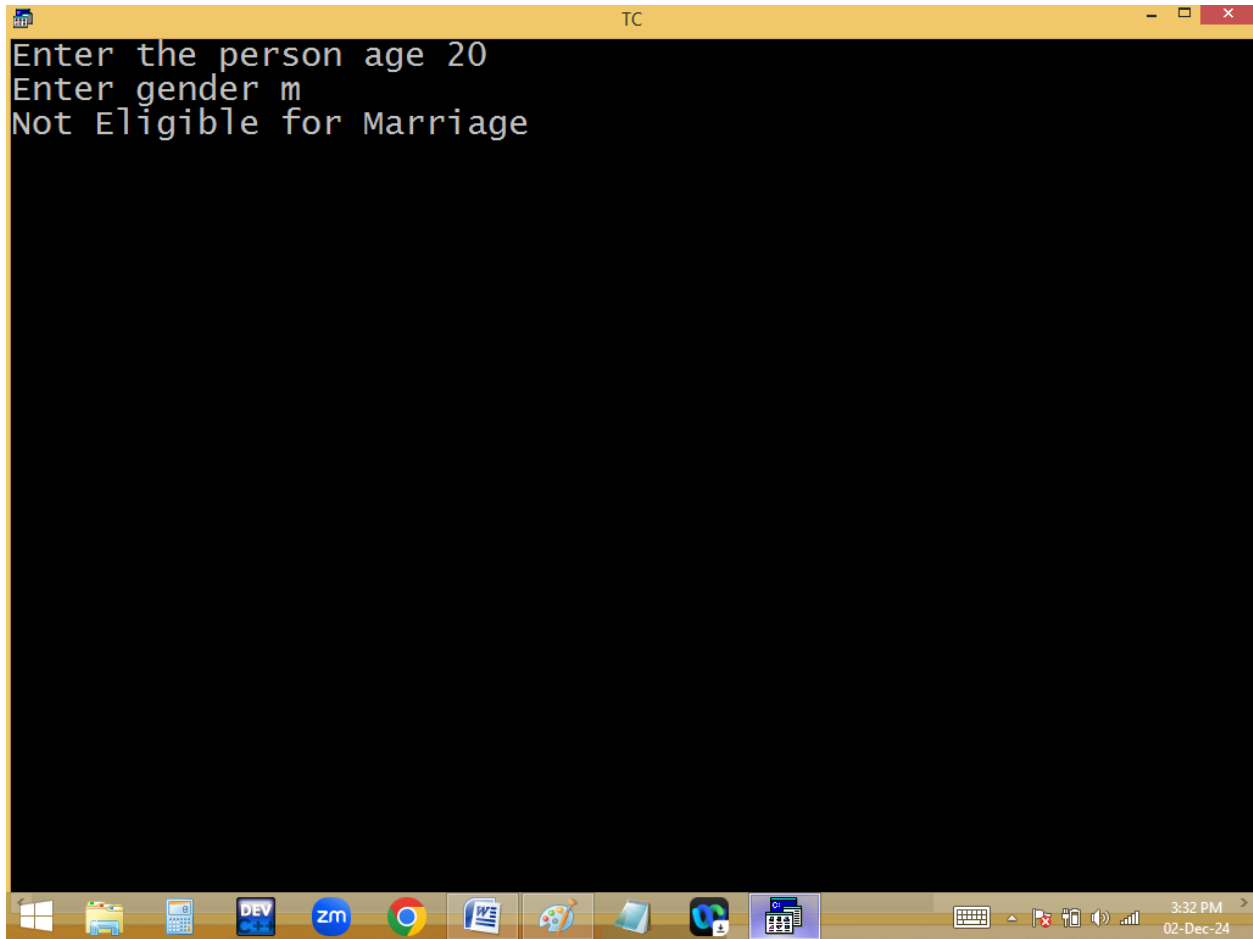
```
TC
File Edit Run Compile Project Options Debug
Line 17 Col 29 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int age;
char gen;
clrscr();
printf("Enter the person age "); scanf("%d",&age);
flushall();
printf("Enter gender "); scanf("%c",&gen);
if(gen=='M' || gen=='m' || gen=='f' || gen=='F')
{
if(age>=21 || (gen=='f' || gen=='F') && age>=18)
puts("Eligible for marriage");
else puts("Not Eligible for Marriage");
}
else puts("Invalid Gender");
getch();
}
```

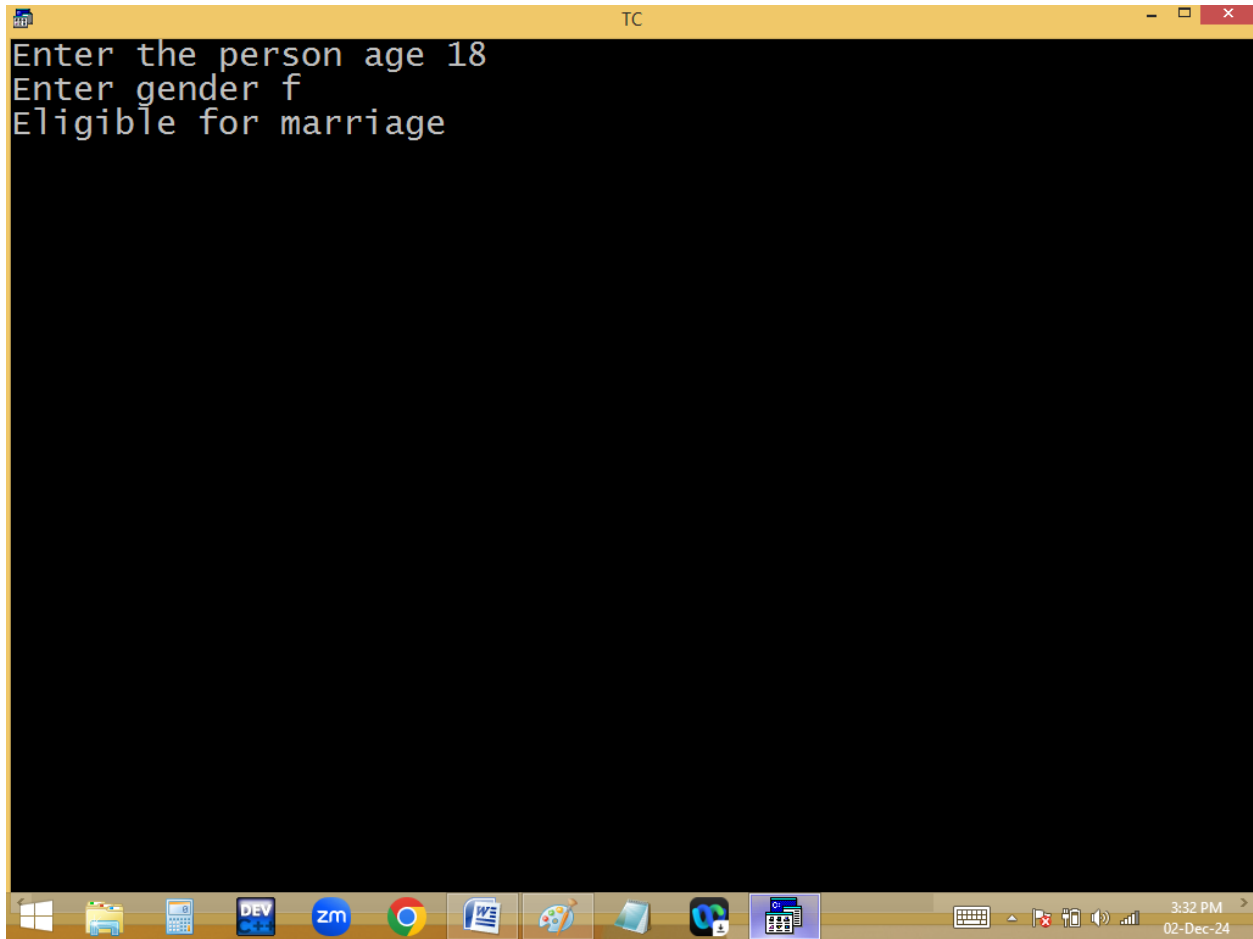
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

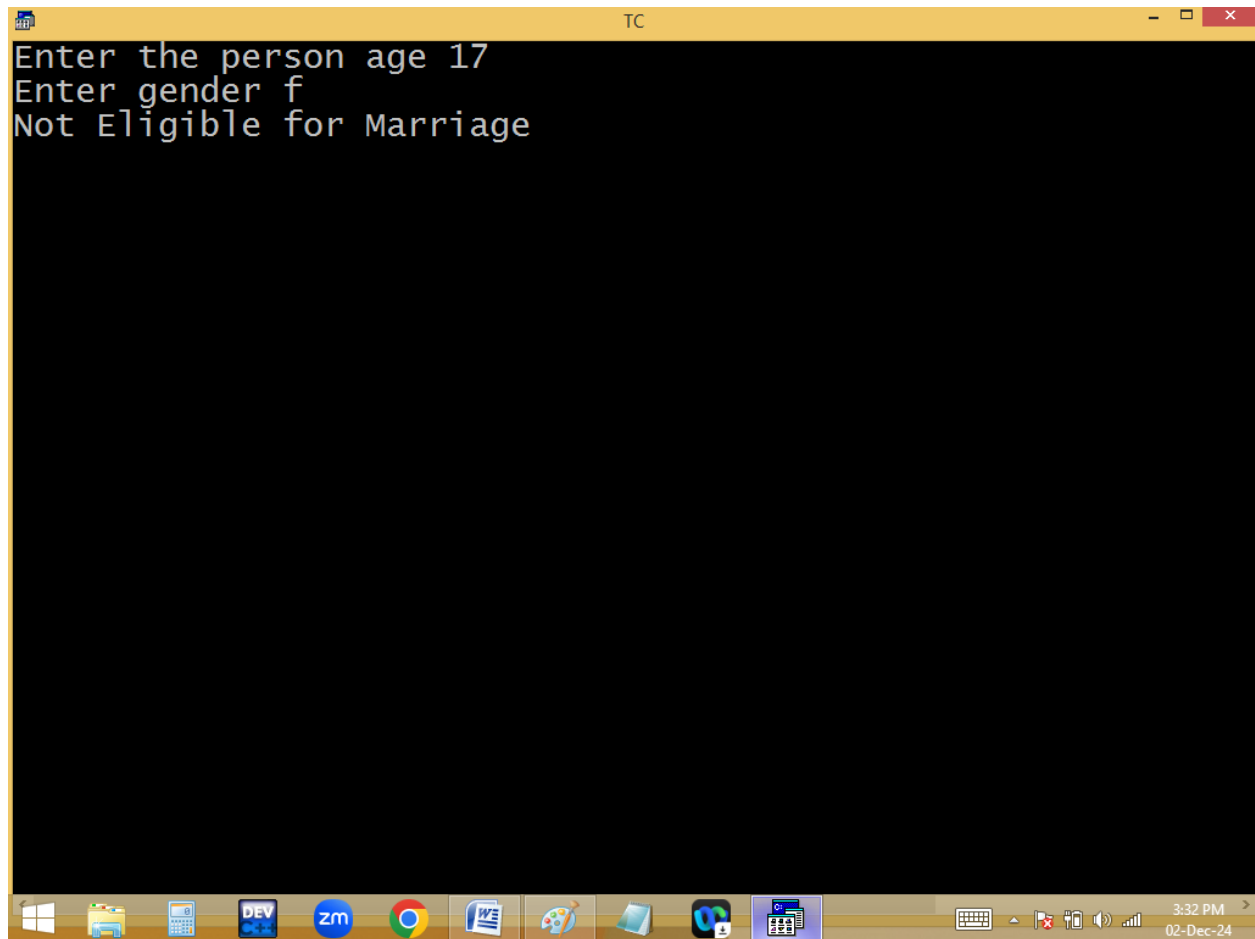
3:32 PM 02-Dec-24



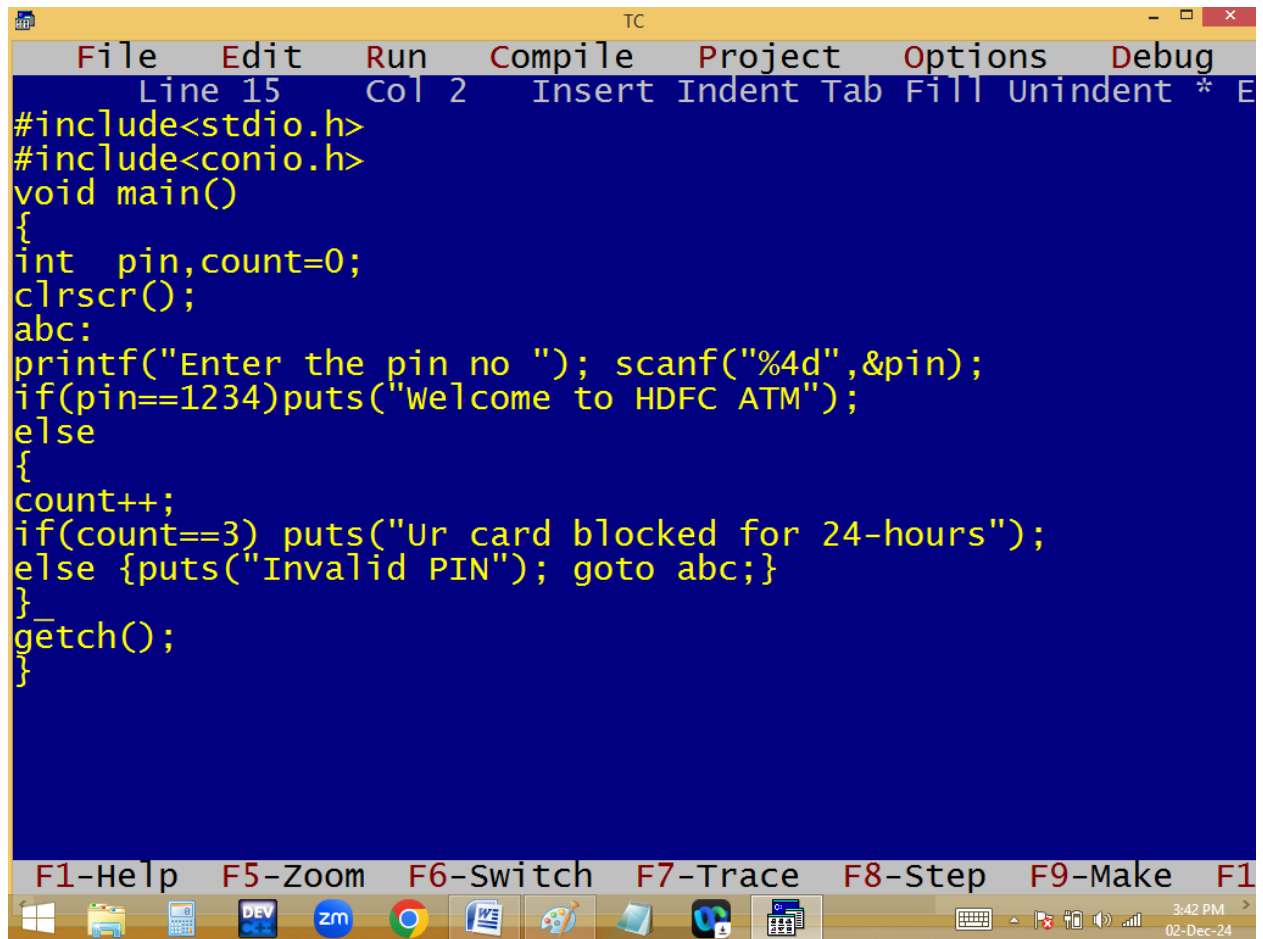








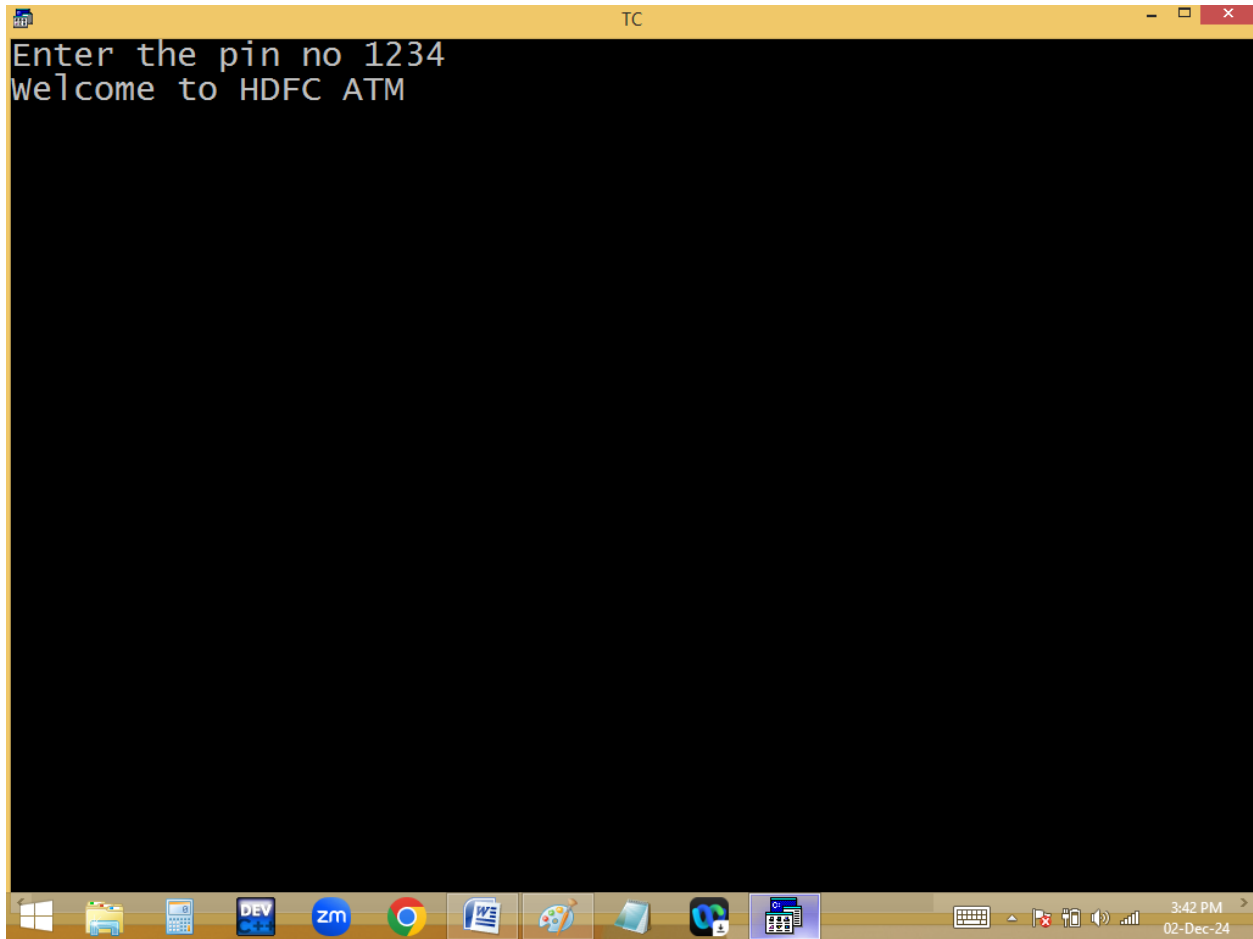
ATM PIN Validation:

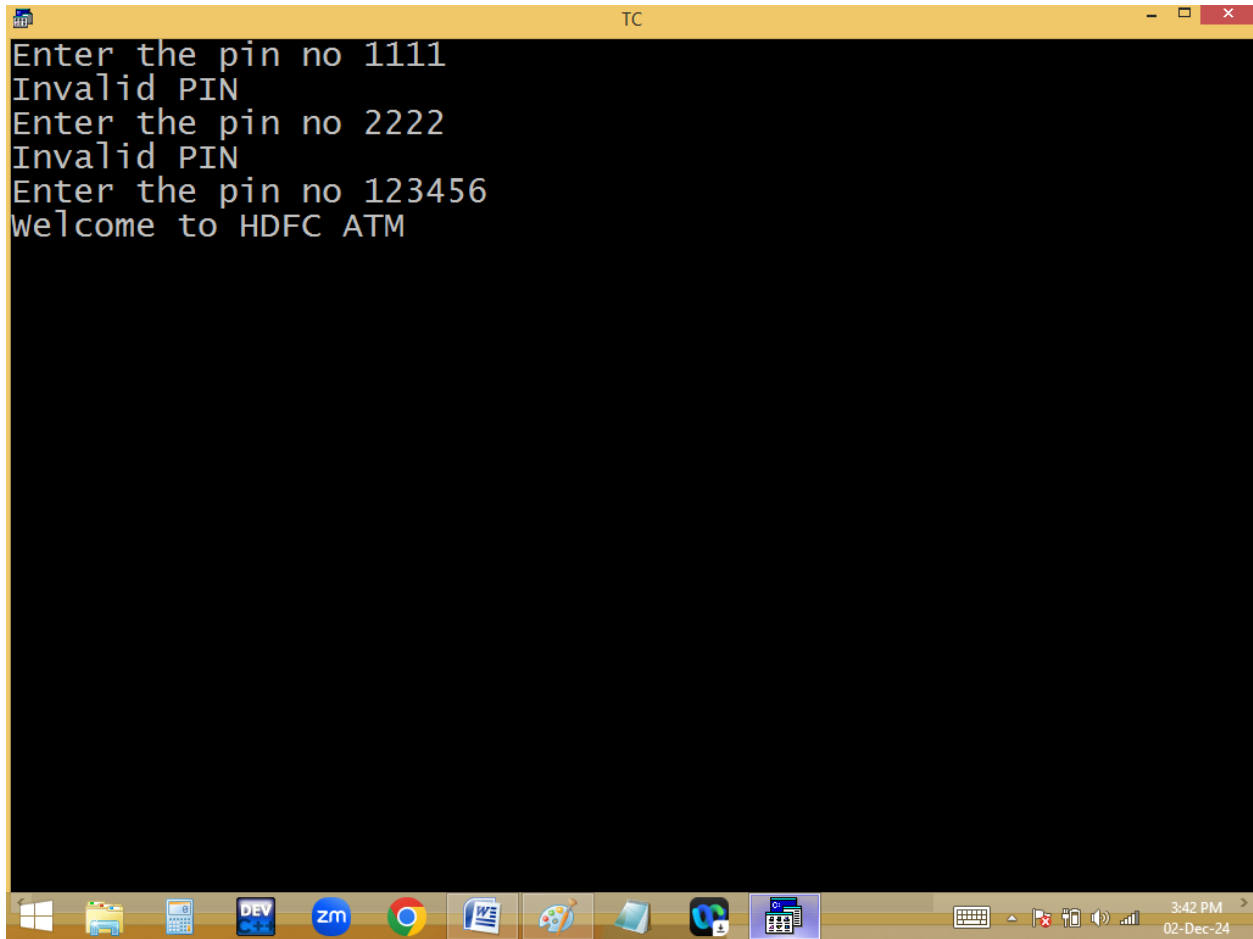


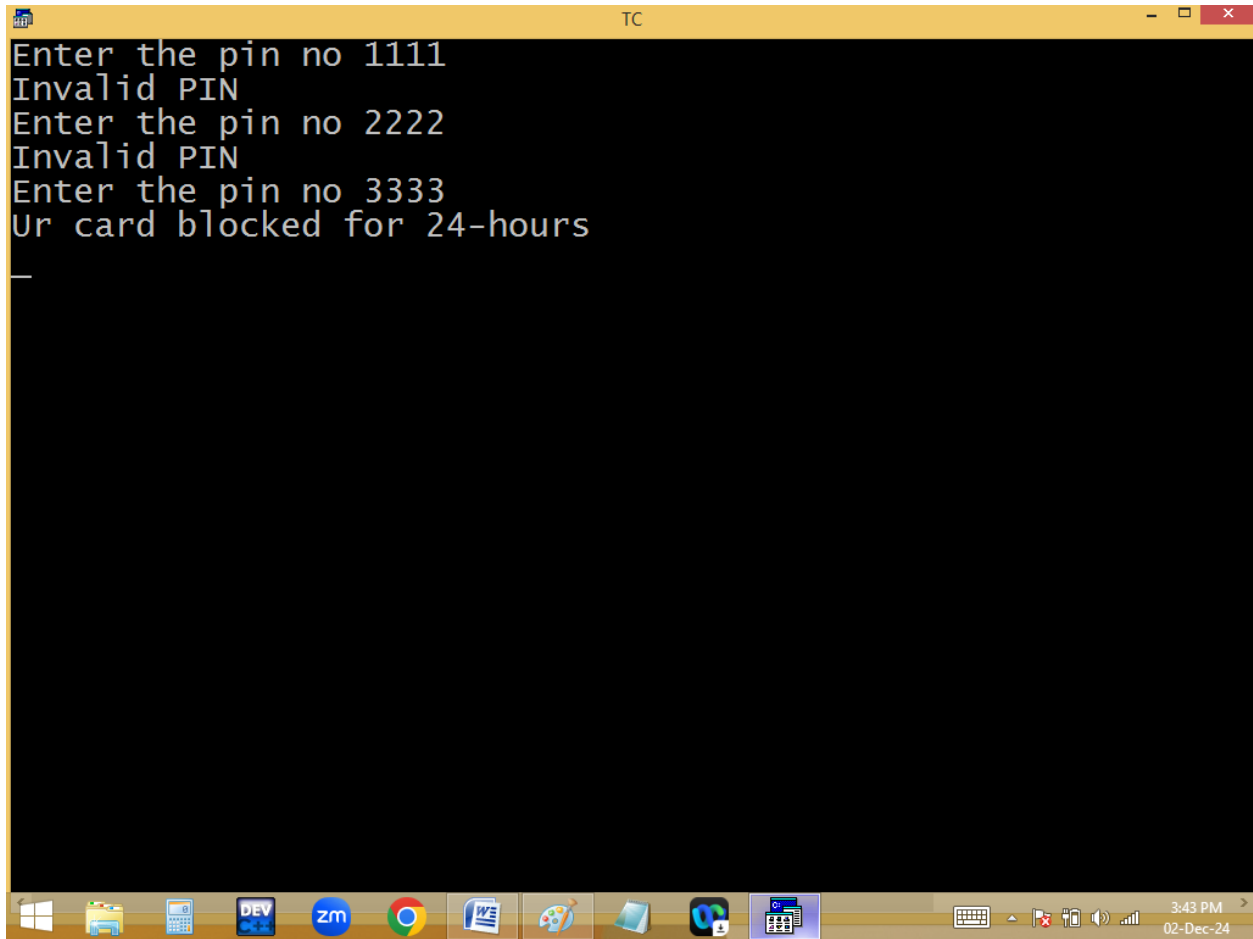
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 15 Col 2" and lists editing actions: Insert, Indent, Tab, Fill, Unindent, and *. The main editing area has a blue background and contains the following C code:

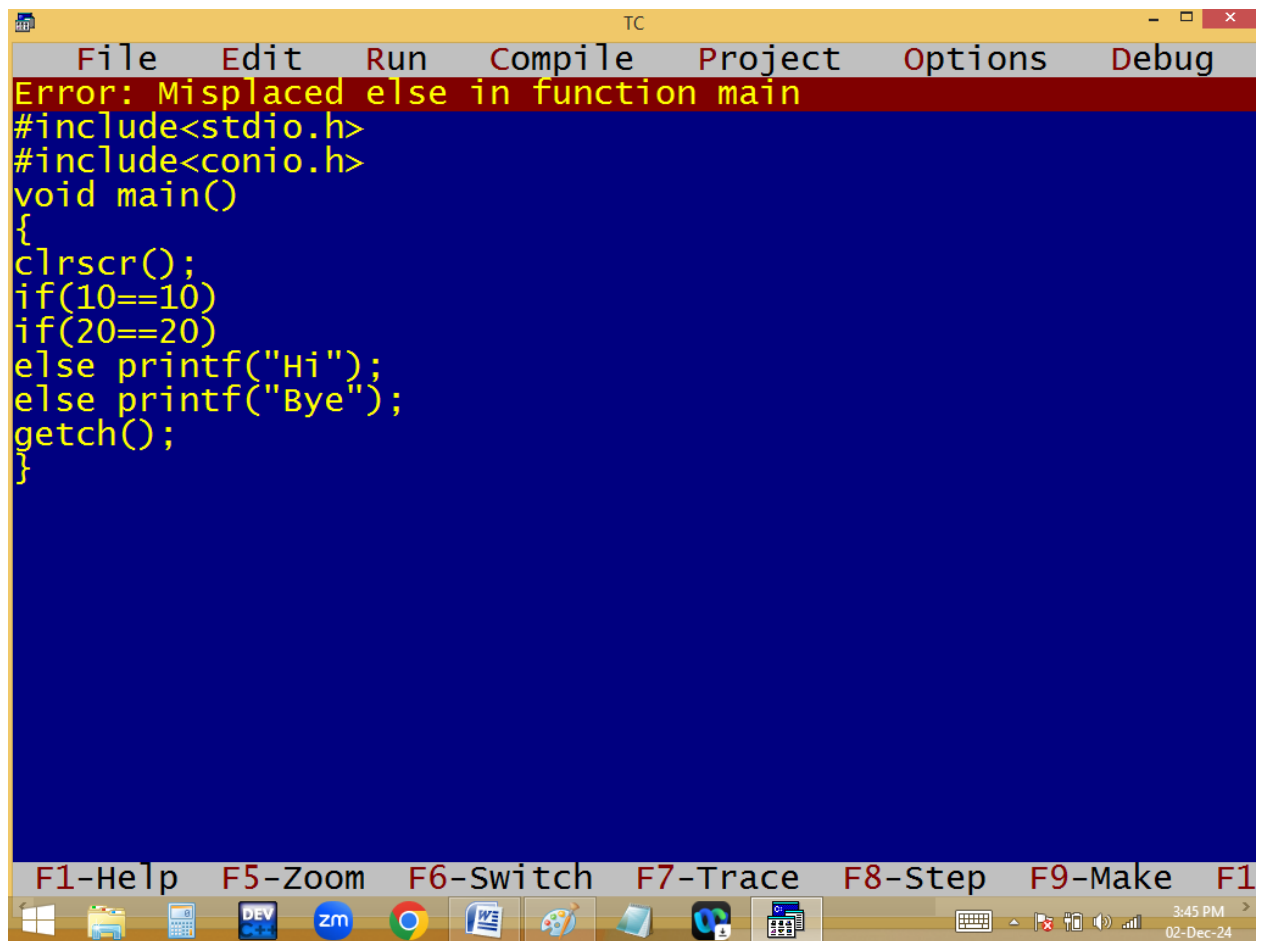
```
#include<stdio.h>
#include<conio.h>
void main()
{
int  pin,count=0;
clrscr();
abc:
printf("Enter the pin no "); scanf("%4d",&pin);
if(pin==1234)puts("Welcome to HDFC ATM");
else
{
count++;
if(count==3) puts("Ur card blocked for 24-hours");
else {puts("Invalid PIN"); goto abc;}}
getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. The Windows taskbar is visible at the bottom, showing icons for File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and VS Code. The system clock in the bottom right corner shows 3:42 PM on 02-Dec-24.





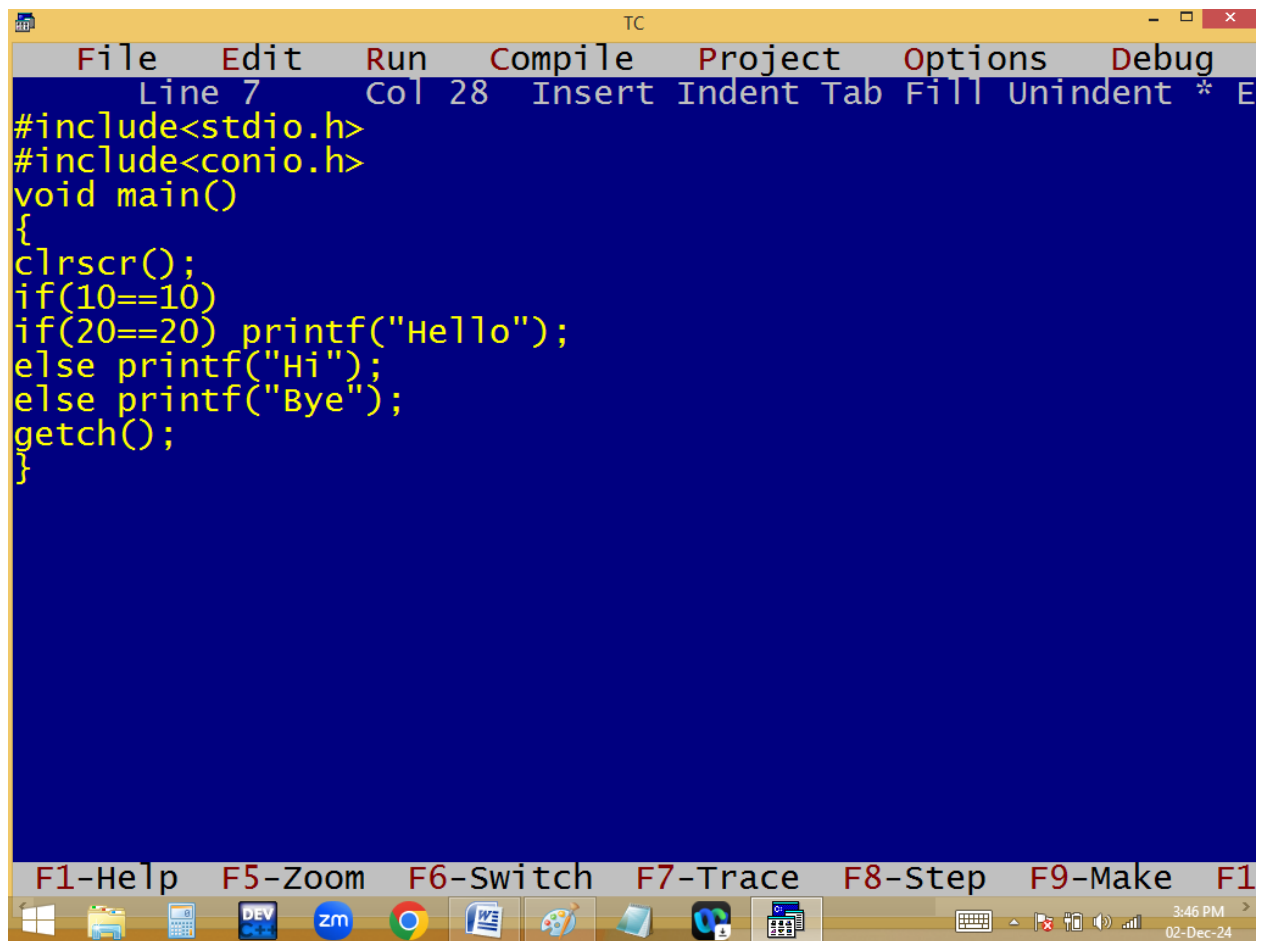




The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the code editor states: "Error: Misplaced else in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(10==10)
if(20==20)
else printf("Hi");
else printf("Bye");
getch();
}
```

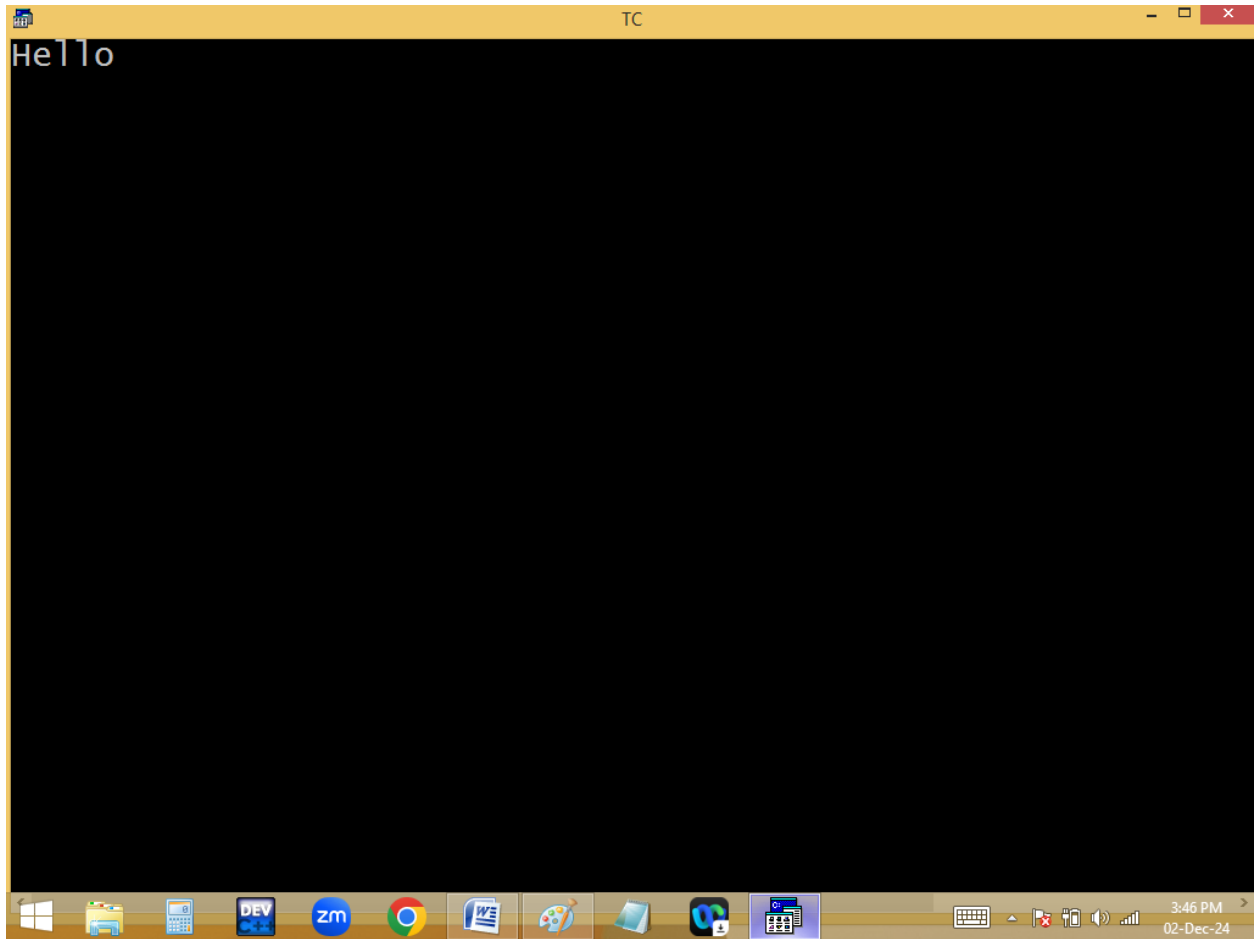
Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the bottom shows various icons including the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder icon. The system tray on the right shows the time as 3:45 PM and the date as 02-Dec-24.

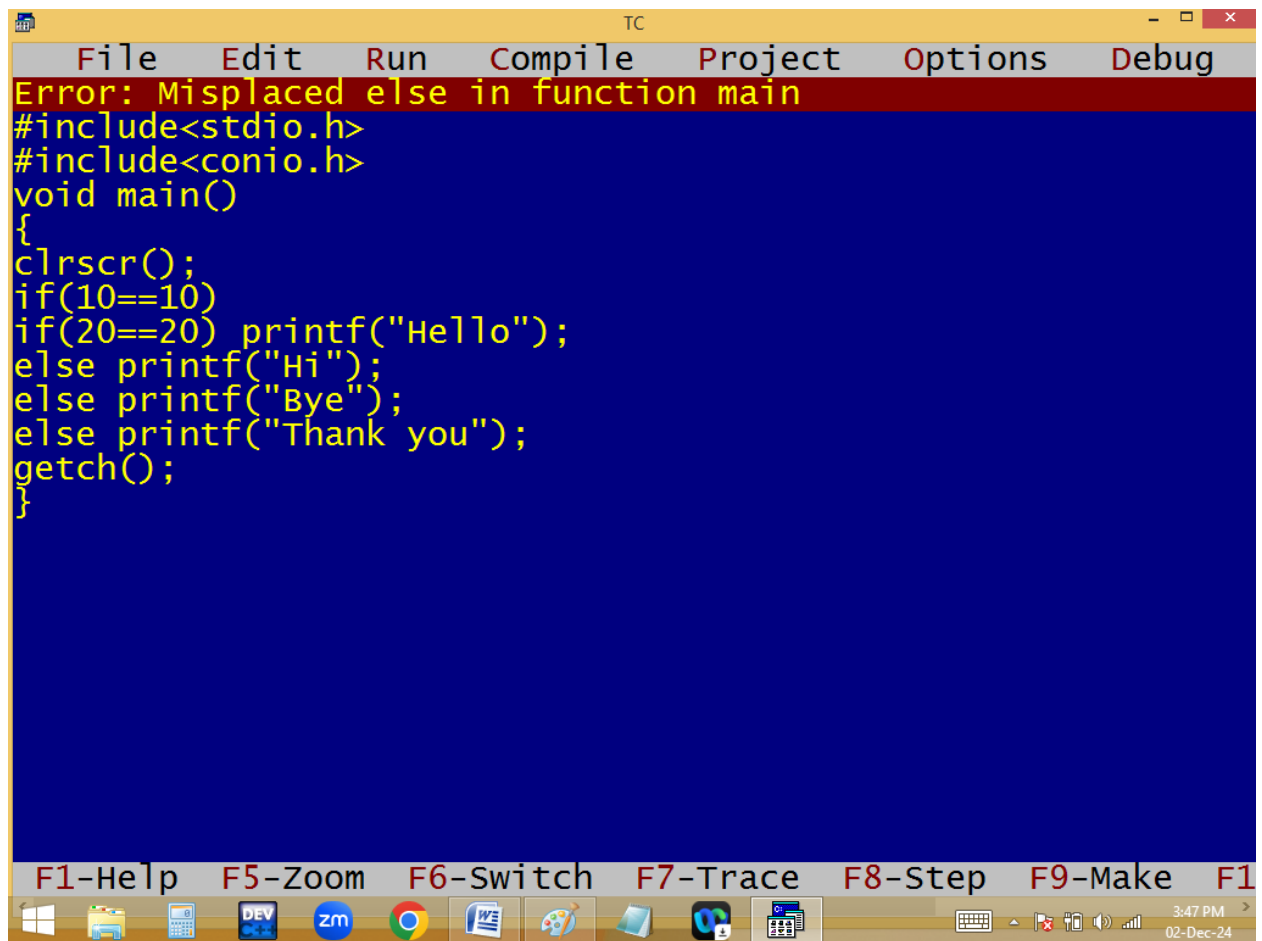


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 7", "Col 28", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and "* E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(10==10)
if(20==20) printf("Hello");
else printf("Hi");
else printf("Bye");
getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and the Turbo C++ application. The system clock in the bottom right corner shows "3:46 PM" and "02-Dec-24".

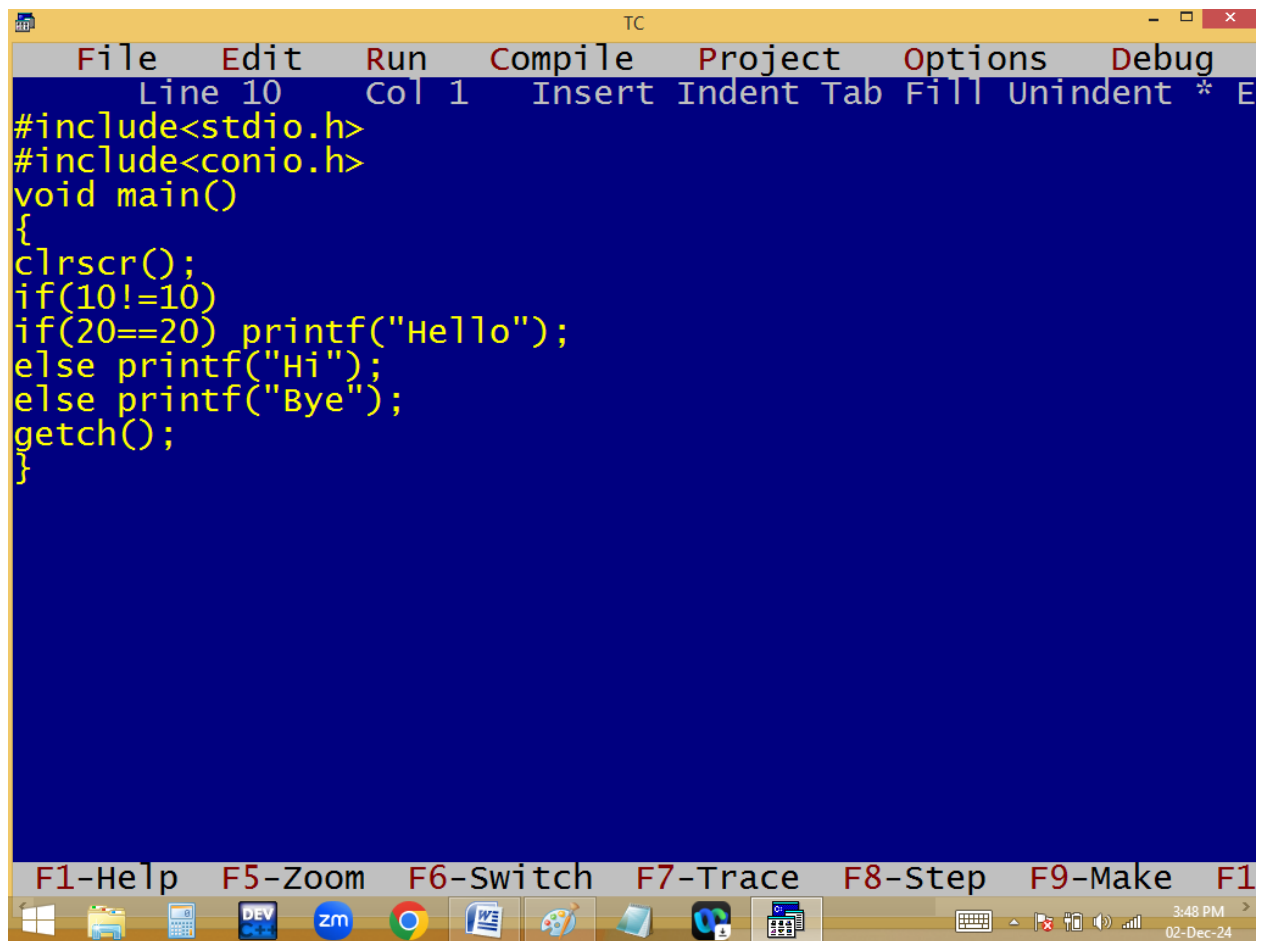




The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the code editor states: "Error: Misplaced else in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(10==10)
if(20==20) printf("Hello");
else printf("Hi");
else printf("Bye");
else printf("Thank you");
getch();
}
```

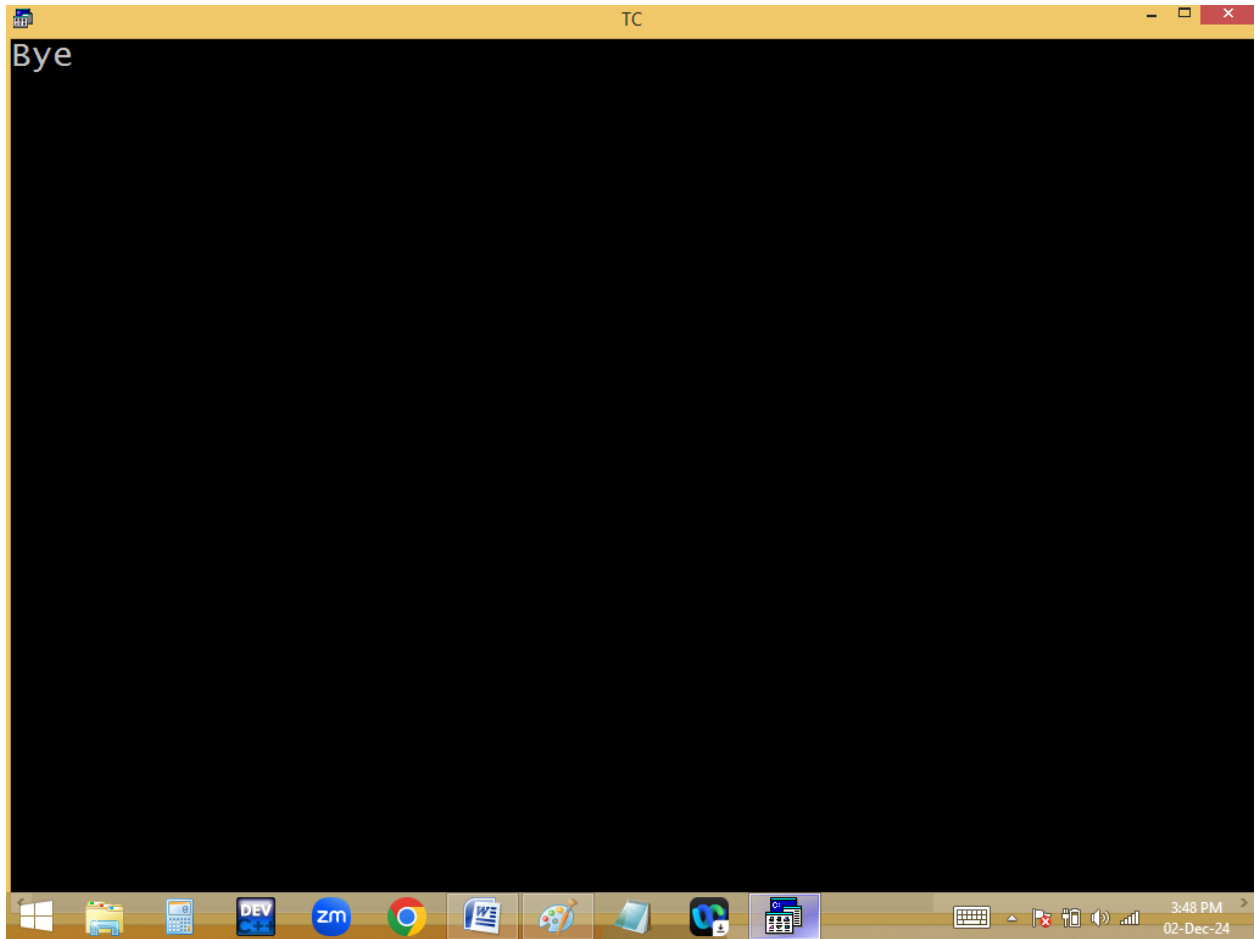
Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the bottom shows various application icons, including the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and others. The system clock in the bottom right corner displays "3:47 PM" and "02-Dec-24".

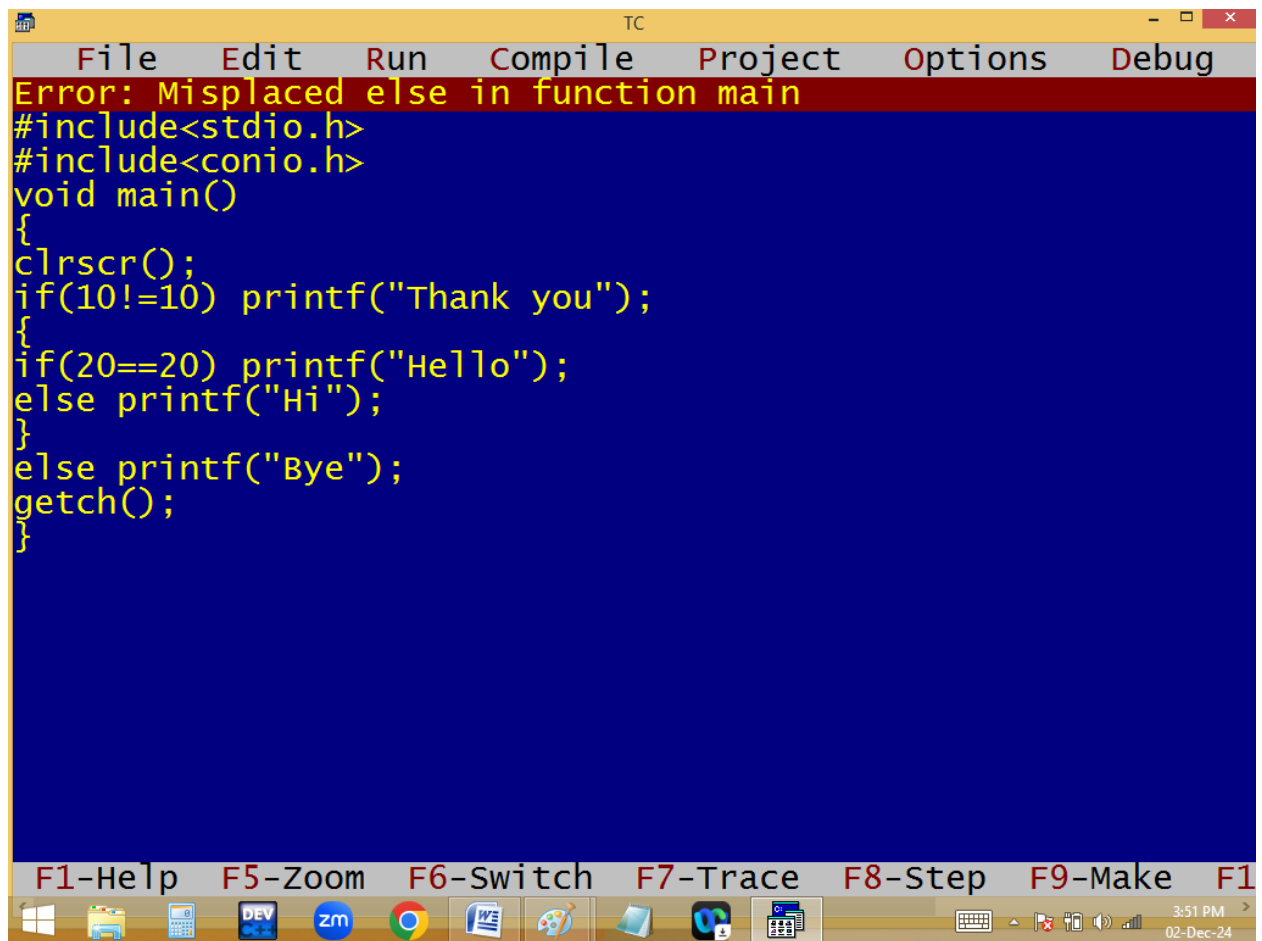


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 10", "Col 1", and lists editing options: "Insert", "Indent", "Tab", "Fill", "Unindent", and "E". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(10!=10)
if(20==20) printf("Hello");
else printf("Hi");
else printf("Bye");
getch();
}
```

The bottom of the window features a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the very bottom shows various application icons and the system clock, which reads "3:48 PM" and "02-Dec-24".





The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the code area states "Error: Misplaced else in function main". The code in the editor is as follows:

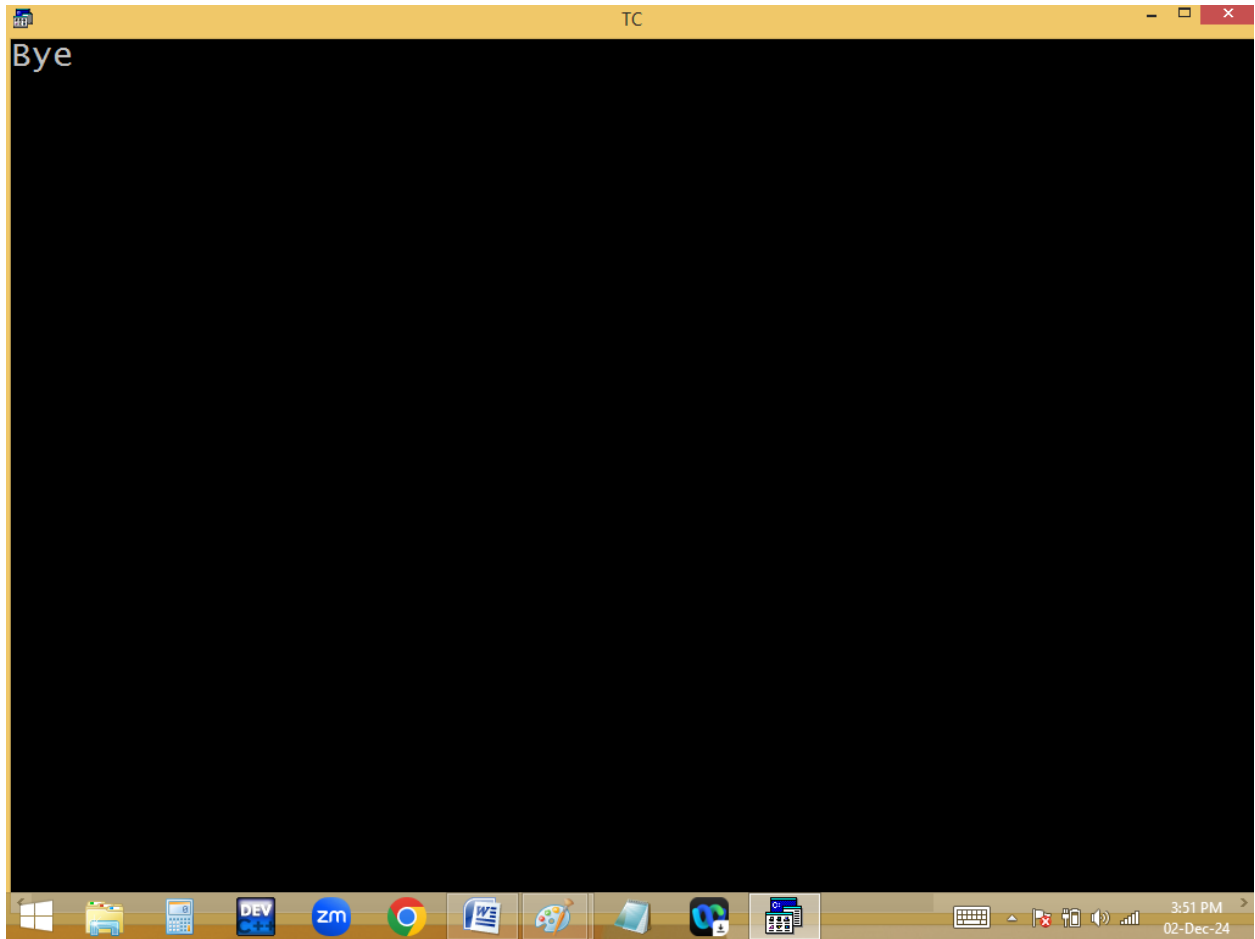
```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(10!=10) printf("Thank you");
}
if(20==20) printf("Hello");
else printf("Hi");
}
else printf("Bye");
getch();
}
```

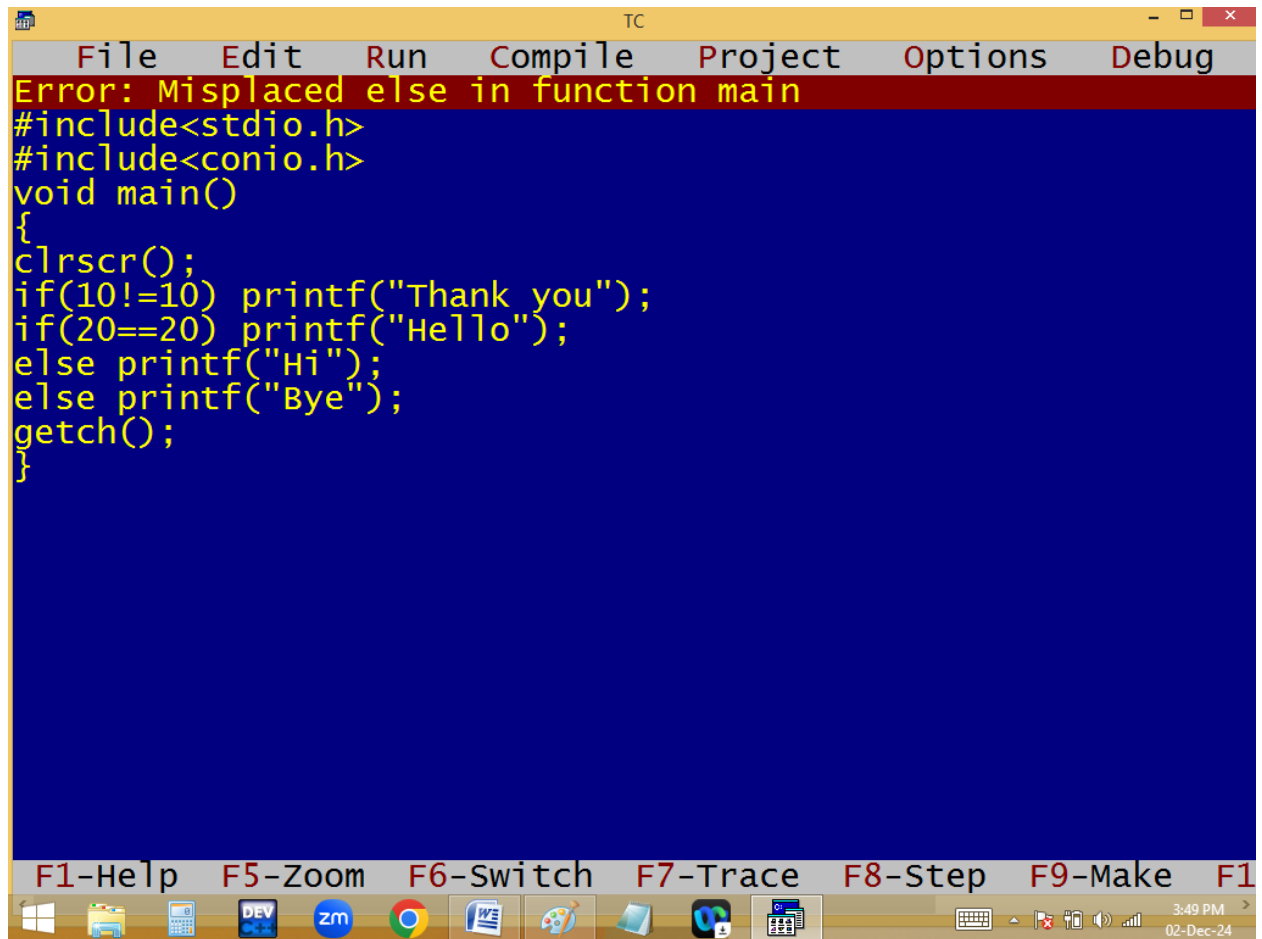
Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the bottom shows various application icons, including the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Notepad, Paint, and several instances of the IDE. The system clock in the bottom right corner displays "3:51 PM" and "02-Dec-24".

```
TC
File Edit Run Compile Project Options Debug
Line 8 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(10!=10)
{
printf("Thank you");
if(20==20) printf("Hello");
else printf("Hi");
}
else printf("Bye");
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

3:51 PM 02-Dec-24





The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the code editor states: "Error: Misplaced else in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(10!=10) printf("Thank you");
if(20==20) printf("Hello");
else printf("Hi");
else printf("Bye");
getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the bottom shows various application icons, including the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and VS Code. The system clock in the bottom right corner displays "3:49 PM" and "02-Dec-24".